# Mobile Application Development Lab

# CSL 341

# Lab Journal 01

## Teacher: Muhammad Athesham Noor



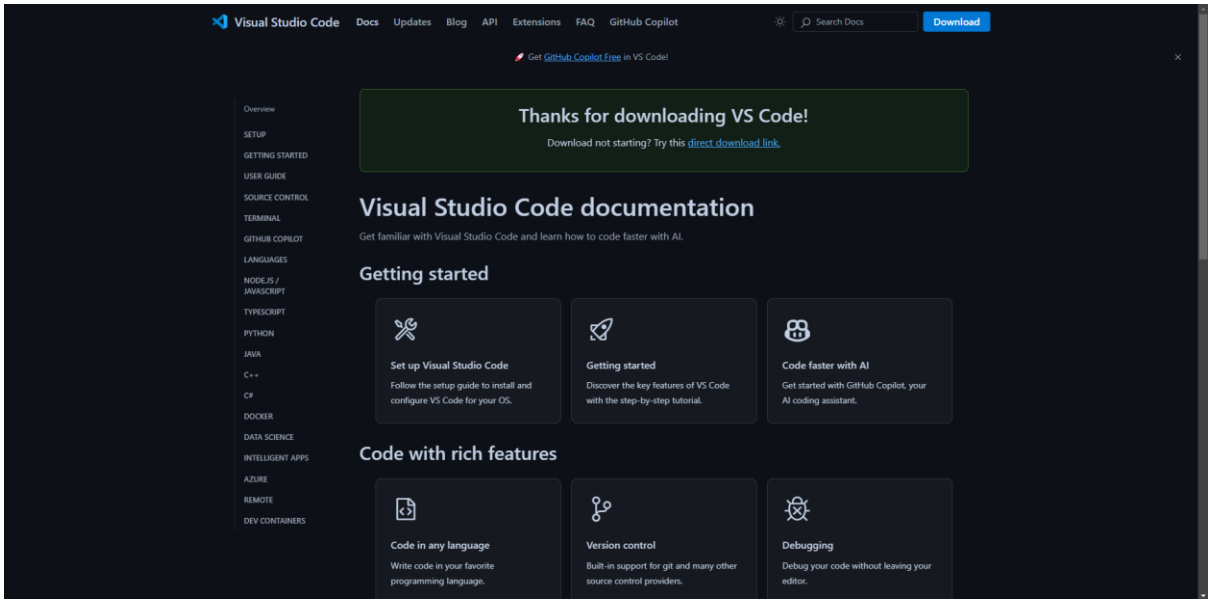

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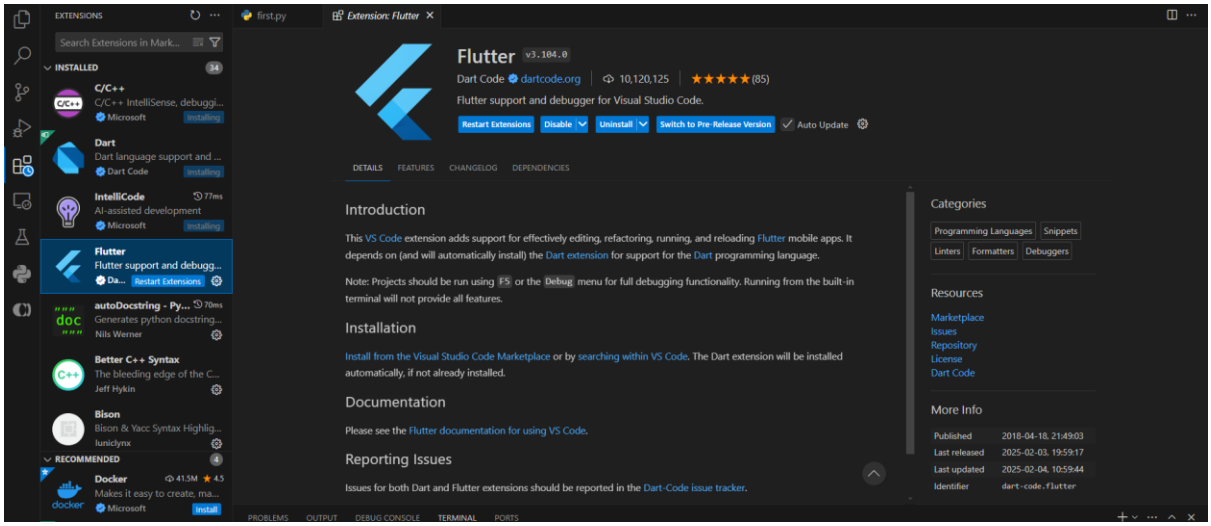
**Question:**

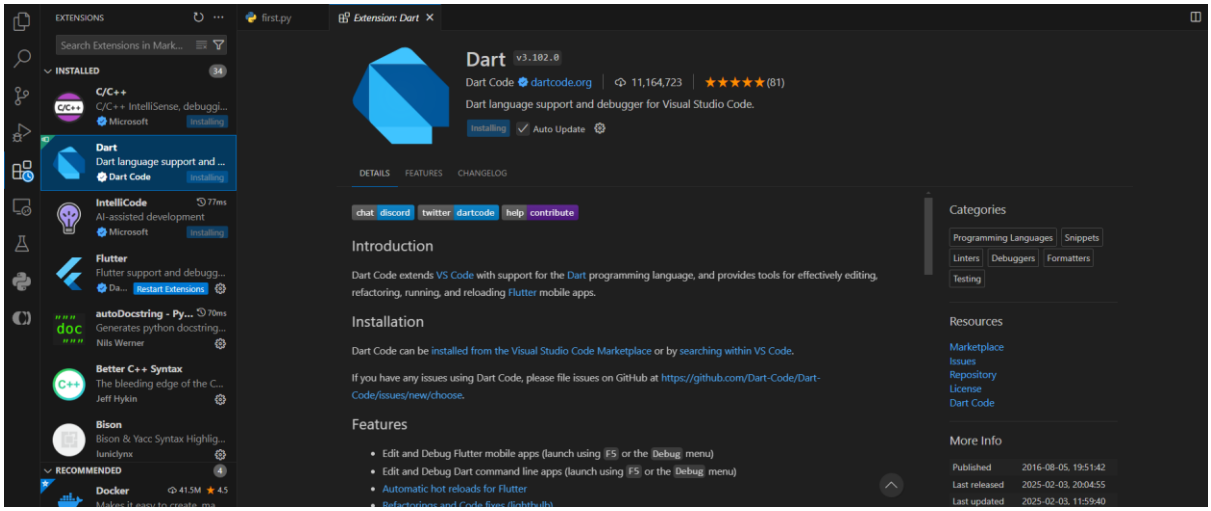
Document the installation process of Visual Studio Code and Flutter SDK on your PC by taking screenshots at each step. These steps should include:

1. Downloading and installing Visual Studio Code.

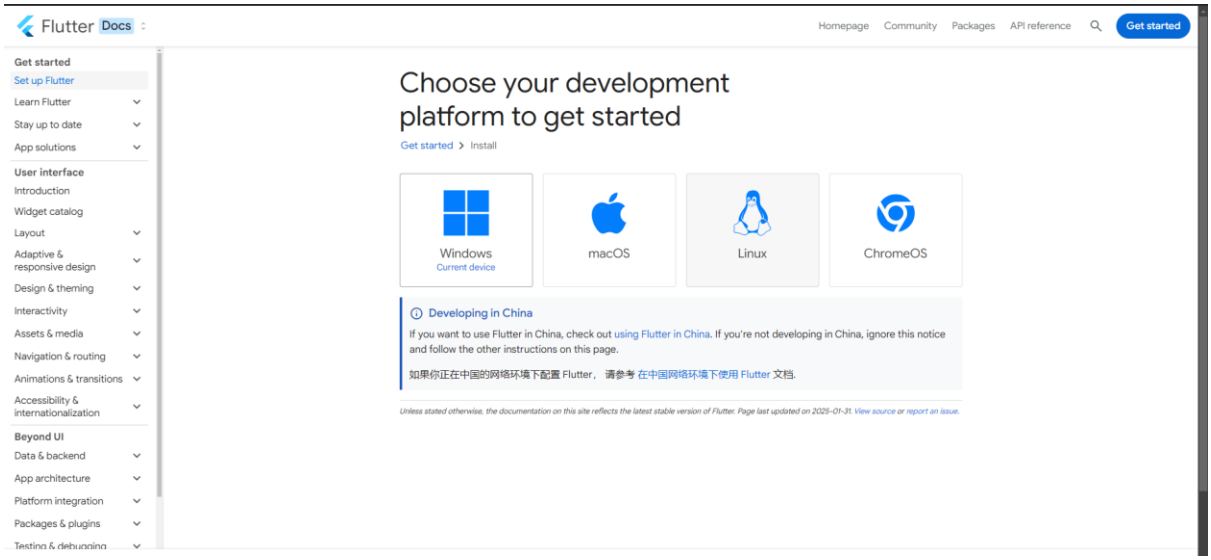


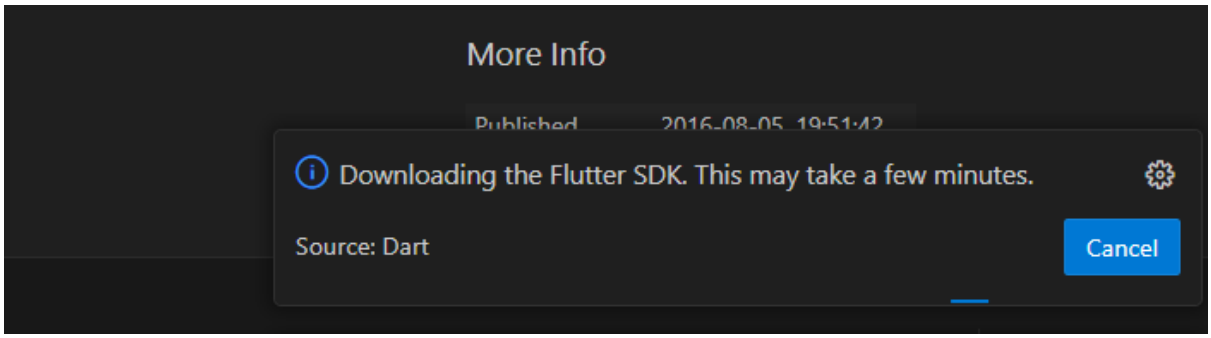

2. Installing the necessary Flutter and Dart extensions in VS Code.



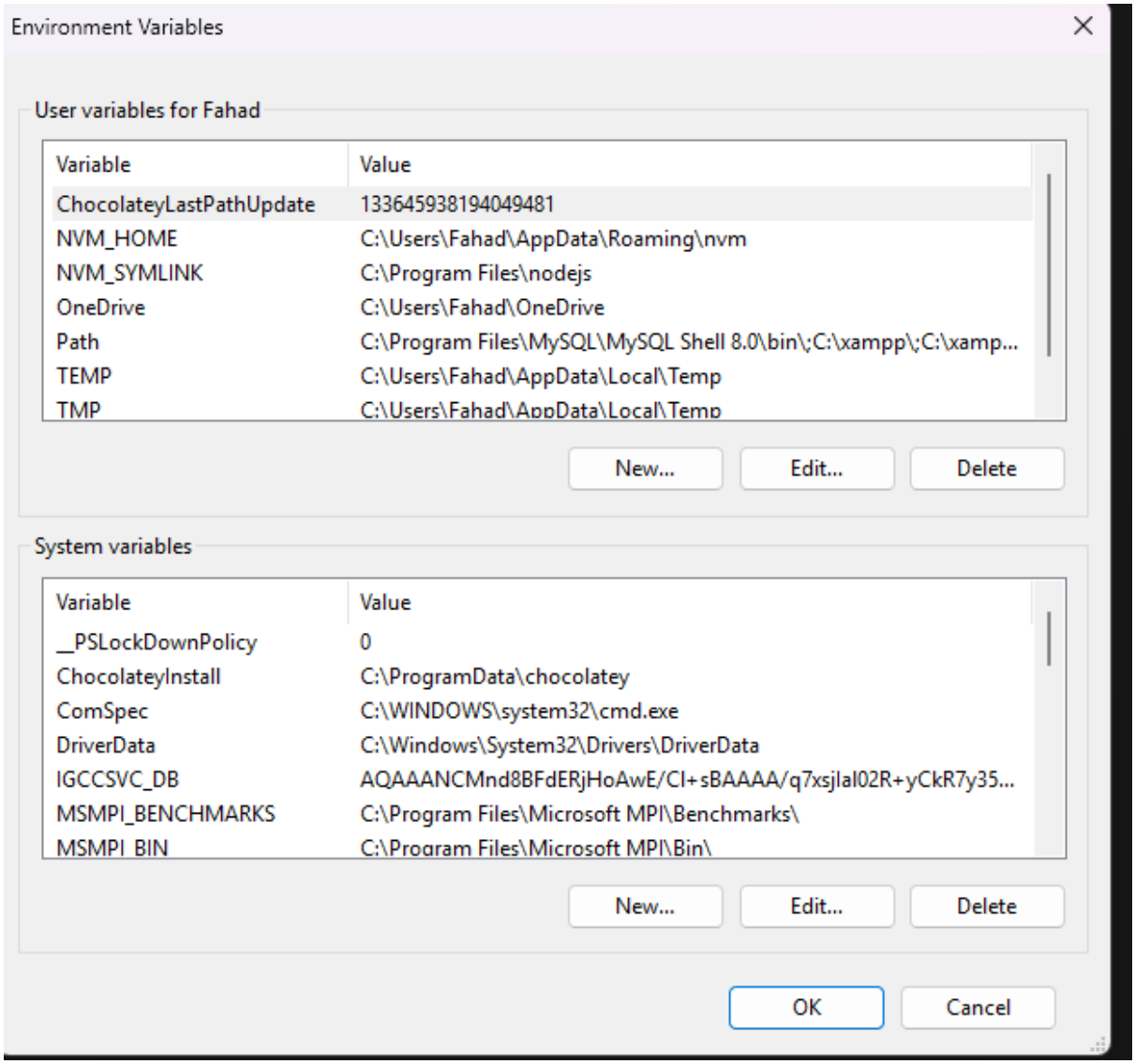



3. Downloading and setting up the Flutter SDK.




4. Configuring the environment variables for Flutter.

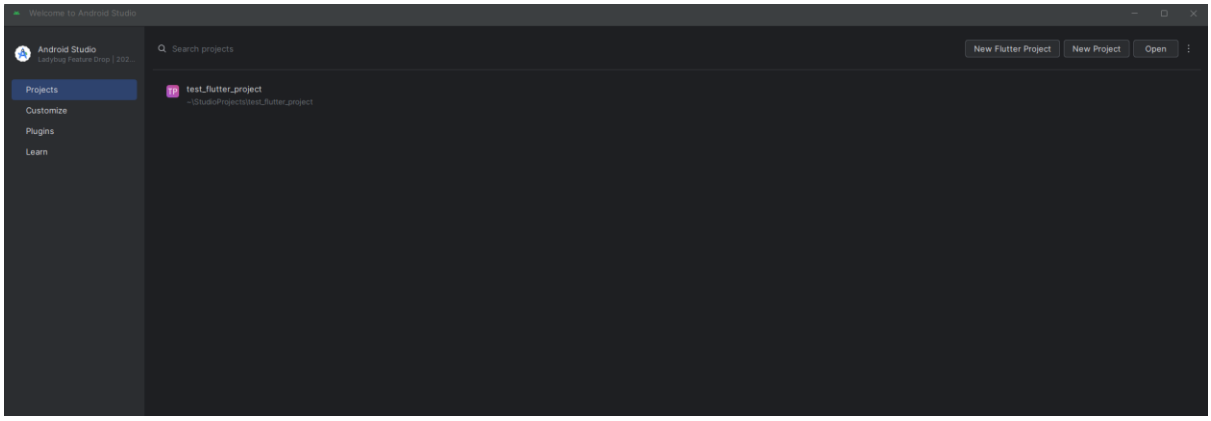



5. Verifying the installation using flutter doctor.

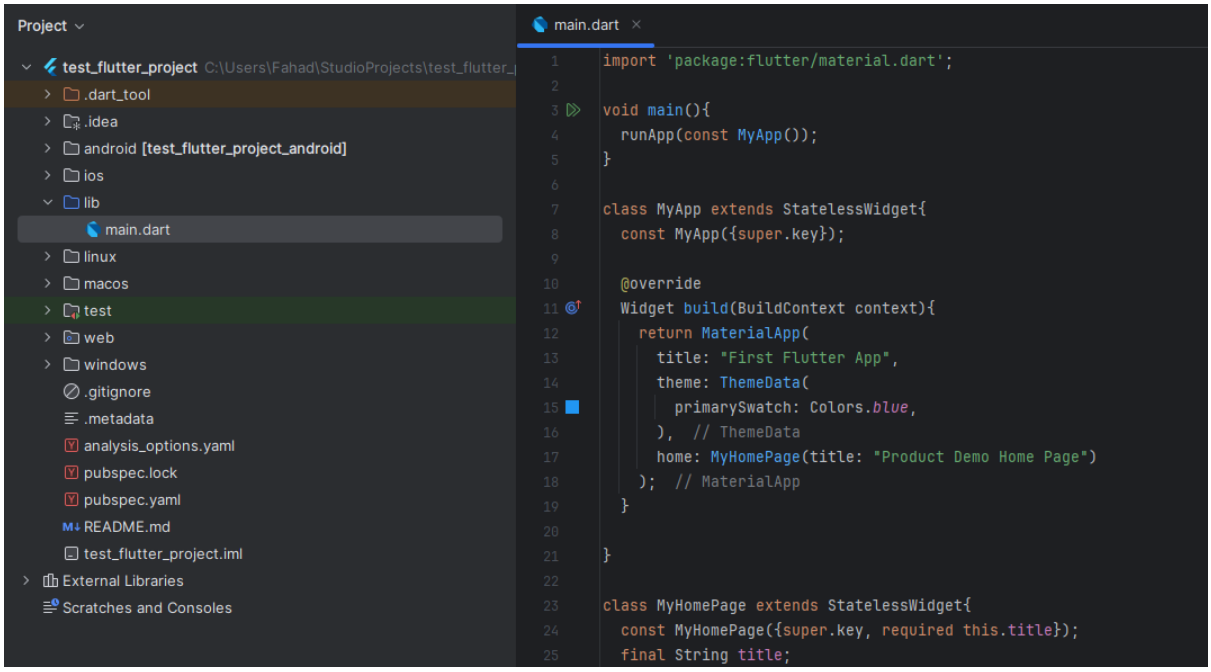
```
PS C:\WINDOWS\system32> flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[√] Flutter (Channel stable, 3.27.3, on Microsoft Windows [Version 10.0.26100.2894], locale en-PK)
[√] Windows Version (Installed version of Windows is version 10 or higher)
[√] Android toolchain - develop for Android devices (Android SDK version 35.0.1)
[√] Chrome - develop for the web
[√] Visual Studio - develop Windows apps (Visual Studio Community 2022 17.7.5)
[√] Android Studio (version 2024.2)
[√] VS Code (version 1.97.1)
[√] Connected device (3 available)
[√] Network resources

• No issues found!
```

6. Install Android Studio (Optional but Recommended for Emulator Support)




7. Configure Flutter with Android Studio ((Optional but Recommended for Emulator Support))




Attach the screenshots of each step to demonstrate the successful setup for Flutter development.

# Mobile Application Development Lab

# CSL 341

# Lab Journal 02

## Teacher: Muhammad Athesham Noor




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**Question:**

Make an app displaying Text and a button on the page.

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MainApp());
}

class MainApp extends StatelessWidget {
  const MainApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        body: Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              Text('I am Evil nooo'),
              ElevatedButton(
                onPressed: () {
                  print("Button pressed");
                },
                child: Text('Elevated Button'),
              ),
            ],
          ),
        ),
      ),
    );
  }
}
```

# Mobile Application Development Lab

# CSL 341

# Lab Journal 03

## Teacher: Muhammad Athesham Noor




Department of Computer Science

BAHRIA UNIVERSITY, ISLAMABAD

# Lab Title:

Building a Basic Mobile Application with Dart and Flutter

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## Objective:

To design and develop a simple mobile application using Dart and Flutter, showcasing basic Flutter components like Scaffold, AppBar, and Text Widgets.

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## Theory:

Flutter is an open-source UI software development kit created by Google. It allows developers to build natively compiled applications for mobile, web, and desktop from a single codebase. This lab focuses on the basics of Flutter, including:

- **Scaffold**: Provides the structure for the app layout.
- **AppBar**: The top section of the app containing the title or action buttons.
- **Text Widget**: A widget to display text in the app.

---

## Apparatus/Materials:

- **Dart SDK**
- **Flutter SDK**
- **Android Studio/VSCode (IDE)**
- **Mobile Emulator or Physical Android/iOS Device**
- **Flutter Packages** (if needed)

---

## Procedure:

1. **Setting up Flutter:**
    - Ensure the Flutter SDK and Dart are installed.
    - Set up the development environment with an IDE (VSCode or Android Studio).
    - Run `flutter doctor` to check the installation and ensure the environment is ready.
2. **Creating a New Flutter Project:**
    - Open the terminal and run `flutter create lab3_mobile_app`.
    - Navigate to the project folder using `cd lab3_mobile_app`.
3. **Implement the UI:**
    - Modify `lib/main.dart` to include the following:
        - Create a basic layout using **Scaffold**.

        - Add an **AppBar** with a title.
        - Use a **Text Widget** to display a welcome message on the app's home screen.
4. **Running the Application:**
    - Run the app on an emulator or connected device using the command `flutter run`.
    - Verify that the app displays the message correctly on the screen.

---

## Code:

```
import 'package:flutter/material.dart';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        appBar: AppBar(
          title: Text('Lab 3 - MAD Course'),
        ),
        body: Center(
          child: Text(
            'Welcome to Flutter!',
            style: TextStyle(fontSize: 24),
          ),
        ),
      ),
    );
  }
}
```

---

## Observations:

1. **Initial App Structure:**
    - The app successfully displayed an AppBar with the title **"Lab 3 - MAD Course"**.
    - The welcome message **"Welcome to Flutter!"** was centered on the screen.
2. **Program Output:**
    - The app ran smoothly on the emulator without errors.
    - The text was properly displayed with a font size of 24.

---

## Results/Analysis:

The basic mobile application was created using Flutter, and it successfully demonstrated the use of key Flutter components like **Scaffold**, **AppBar**, and **Text Widgets**. The application ran correctly and displayed the expected outputs.

---

## Conclusion:

This lab introduced the core components of Flutter, including the basic app structure and UI components. It was a successful demonstration of how to build a simple mobile application with a clean layout. The project highlighted the ease of using Flutter for rapid mobile app development.

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## Future Work/Improvements:

To further enhance this application, we could:

- Add more widgets like buttons, images, and input fields.
- Implement state management to handle user interactions.
- Explore layout widgets like **Row** and **Column** to organize content more effectively.

---

## References:


1. Flutter Documentation - https://flutter.dev/docs
2. Dart Language Overview - https://dart.dev/guides
3. Stack Overflow (for troubleshooting tips) - https://stackoverflow.com/


## 1. Write a program to print your name in Dart.

```
void main() {
  print("Muhammad Ahtesham Noor");
}
```

**Code:**

```
void main() {
  print("My name is Khawaja Fahad Zia");
}
```

**Output:**

```
My name is Khawaja Fahad Zia
```

---

## 2. Write a program to print: Hello I am “John Doe” and Hello I’am “John Doe” using single and double quotes.

```
void main() {
  print('Hello I am "John Doe"');
  print("Hello I'am 'John Doe'");
}
```

**Code:**

```
void main() {
  print('Hello I am "John Doe"');
  print("Hello I am 'John Doe'");
}
```

**Output:**

```
Hello I am "John Doe"
Hello I am 'John Doe'
```

---

## 3. Declare a constant of type int and set its value to 7.

```
void main() {
  const int value = 7;
  print(value);
}
```

**Code:**

```
void main() {
  const int value = 7;
  print(value);
}
```

**Output:**

```
7
```

---

## 4. Write a program in Dart that finds simple interest. Formula = (p * t * r) / 100

```
void main() {
  double p = 1000; // Principal amount
  double t = 2;    // Time in years
  double r = 5;    // Rate of interest

  double simpleInterest = (p * t * r) / 100;
  print("Simple Interest: $simpleInterest");
}
```

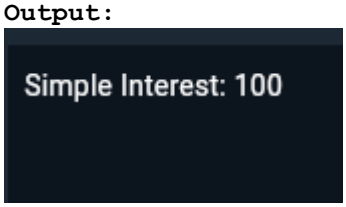
**Code:**

```
void main() {
  double p = 1000; // Principal amount
  double t = 2;    // Time in years
  double r = 5;    // Rate of interest

  double simpleInterest = (p * t * r) / 100;
```

```
  print("Simple Interest: $simpleInterest");
}
```

**Output:**




---

## 5. Write a program to print the square of a number using user input.

```
import 'dart:io';

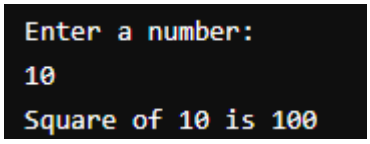
void main() {
  print('Enter a number:');
  int number = int.parse(stdin.readLineSync()!);
  print('Square of $number is ${number * number}');
}
```

**Code:**

```
import 'dart:io';

void main() {
  print('Enter a number:');
  int number = int.parse(stdin.readLineSync()!);
  print('Square of $number is ${number * number}');
}
```

**Output:**




---

## 6. Write a program to print the full name from the first name and last name using user input.

```
import 'dart:io';

void main() {
  print('Enter your first name:');
  String firstName = stdin.readLineSync()!;

  print('Enter your last name:');
  String lastName = stdin.readLineSync()!;

  print('Full Name: $firstName $lastName');
}
```

**Code:**

```
import 'dart:io';
```

```
void main() {
  print('Enter your first name:');
  String firstName = stdin.readLineSync()!;

  print('Enter your last name:');
  String lastName = stdin.readLineSync()!;

  print('Full Name: $firstName $lastName');
}
```

**Output:**

```
Enter your first name:
Fahad
Enter your last name:
Zia
Full Name: Fahad Zia
```

---

## 7. Write a program to find the quotient and remainder of two integers.

```
import 'dart:io';

void main() {
  print('Enter first number:');
  int num1 = int.parse(stdin.readLineSync()!);

  print('Enter second number:');
  int num2 = int.parse(stdin.readLineSync()!);

  int quotient = num1 ~/ num2;
  int remainder = num1 % num2;

  print('Quotient: $quotient');
  print('Remainder: $remainder');
}
```

**Code:**

```
import 'dart:io';

void main() {
  print('Enter first number:');
  int num1 = int.parse(stdin.readLineSync()!);

  print('Enter second number:');
  int num2 = int.parse(stdin.readLineSync()!);

  int quotient = num1 ~/ num2;
  int remainder = num1 % num2;

  print('Quotient: $quotient');
  print('Remainder: $remainder');
}
```

**Output:**

```
Enter first number:
36
Enter second number:
5
Quotient: 7
Remainder: 1
```

---

## 8. Write a program to swap two numbers.

```
void main() {
  int a = 5;
  int b = 10;

  print('Before swap: a = $a, b = $b');

  // Swapping logic
  int temp = a;
  a = b;
  b = temp;

  print('After swap: a = $a, b = $b');
}
```

**Code:**

```
void main() {
  int a = 5;
  int b = 10;

  print('Before swap: a = $a, b = $b');

  // Swapping logic
  int temp = a;
  a = b;
  b = temp;

  print('After swap: a = $a, b = $b');
}
```

**Output:**

```
Before swap: a = 5, b = 10
After swap: a = 10, b = 5
```

---

## 9. Write a program in Dart to remove all whitespaces from a String.

```
void main() {
  String str = "Hello World ";
  String noWhitespaceStr = str.replaceAll(' ', '');
  print(noWhitespaceStr);
}
```

**Code:**

```
void main() {
```

```
  String str = "Hello World ";
  String noWhitespaceStr = str.replaceAll(' ', '');
  print(noWhitespaceStr);
}
```

**Output:**

```
HelloWorld
```

---

## 10. Write a Dart program to convert a String to int.

```
void main() {
  String numberStr = "1234";
  int number = int.parse(numberStr);
  print("Converted number is $number");
}
```

**Code:**

```
void main() {
  String numberStr = "1234";
  int number = int.parse(numberStr);
  print("Converted number is $number");
}
```

**Output:**

```
Converted number is 1234
```

---

## 11. Suppose you often go to restaurants with friends, and you have to split the bill. Write a program to calculate the split amount of the bill. Formula = (total bill amount) / number of people

```
import 'dart:io';

void main() {
  print('Enter the total bill amount:');
  double totalBill = double.parse(stdin.readLineSync()!);

  print('Enter the number of people:');
  int numberOfPeople = int.parse(stdin.readLineSync()!);

  double splitAmount = totalBill / numberOfPeople;
  print('Each person needs to pay: \$${splitAmount.toStringAsFixed(2)}');
}
```

**Code:**

```
import 'dart:io';

void main() {
  print('Enter the total bill amount:');
  double totalBill = double.parse(stdin.readLineSync()!);

  print('Enter the number of people:');
  int numberOfPeople = int.parse(stdin.readLineSync()!);
```

```
  double splitAmount = totalBill / numberOfPeople;
  print('Each person needs to pay: \$${splitAmount.toStringAsFixed(2)}');
}
```

**Output:**

```
Enter the total bill amount:
1000
Enter the number of people:
4
Each person needs to pay: $250.00
```

---

## 12. Suppose your distance to the office from home is 25 km, and you travel at 40 km per hour. Write a program to calculate the time taken to reach the office in minutes. Formula = (distance) / (speed)

```
void main() {
  double distance = 25; // Distance in kilometers
  double speed = 40;    // Speed in kilometers per hour

  double timeInHours = distance / speed;
  double timeInMinutes = timeInHours * 60;

  print("Time to reach the office: ${timeInMinutes.toStringAsFixed(2)}
minutes");
}
```

**Code:**

```
void main() {
  double distance = 25; // Distance in kilometers
  double speed = 40;    // Speed in kilometers per hour

  double timeInHours = distance / speed;
  double timeInMinutes = timeInHours * 60;

  print("Time to reach the office: ${timeInMinutes.toStringAsFixed(2)}
minutes");
}
```

**Output:**

```
Time to reach the office: 37.50 minutes
```

**Question:**

Make an app displaying Text and a button on the page and also add a picture from the assets folder after creating it in the project directory.

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MainApp());
}

class MainApp extends StatelessWidget {
  const MainApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Scaffold(
        body: Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              Container(
                constraints: BoxConstraints(
                  maxWidth: 200,
                  maxHeight: 200,
                ),

                child: GestureDetector(
            onTap: () {
              print('Image tapped!');
            },
            child: Image.asset("assets/Doof.jpg")
                ),
              ),
              Text('I am Evil nooo'),
              ElevatedButton(
                onPressed: () {
                  print("Button pressed");
                },
                child: Text('Elevated Button'),
              ),
            ],
          ),
        ),
      ),
    );
  }
}
```

**Question:**

Use multiple types of buttons in the previous page.

**With Buttons main.dart:**

```dart
import 'package:flutter/material.dart';
import 'package:flutter_application_1/HomePage.dart';
void main() {
  runApp(const MainApp());
}

class MainApp extends StatelessWidget {
  const MainApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: HomePage(),
    );
  }
}
```

**With Buttons HomePage.dart:**

```dart
import 'package:flutter/material.dart';

class HomePage extends StatelessWidget {
  const HomePage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text("I am Doofed out of my mind"),
      centerTitle: true,
       backgroundColor: Colors.black,
        titleTextStyle: TextStyle(color: Colors.white,),
        ),
        floatingActionButton: FloatingActionButton(
          onPressed: () {
            print("Whigga please");
          },
                child: const Icon(Icons.navigation),
```

```
            ),
        body: Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              Container(
                constraints: BoxConstraints(
                  maxWidth: 200,
                  maxHeight: 200,
                ),

                child: GestureDetector(
            onTap: () {
              print('Image tapped!');
            },
            child: Image.asset("assets/Doof.jpg")
                ),
              ),
              Text('I am Evil nooo'),
              ElevatedButton(
                style: ElevatedButton.styleFrom(foregroundColor: Colors.red,),
                onPressed: () {
                  print("Button pressed");
                },
                child: Text('Elevated Button'),
              ),
              OutlinedButton(
                onPressed: () {
                  print("trash");
                },
                child: const Text("This is an OutlinedButton"),
              ),
              TextButton(
                style: TextButton.styleFrom(
                  minimumSize: Size(100,50),
                  backgroundColor: Color.fromARGB(202, 52, 194, 0),
                  foregroundColor: Color.fromARGB(192, 143, 54, 54),
                  ),
                onPressed: () {
                  print("This is a text Button");
                },
                child: const Text("This is a Text Button"),
              ),
            ],
          ),
        ),
      );
  }
```

```
}
```

# Mobile Application Development Lab

# CSL 341

# Lab Journal 4

## Teacher: Muhammad Athesham Noor




Department of Computer Science

BAHRIA UNIVERSITY, ISLAMABAD

**Question:**

Make a simple fast food restaurant index page.

Main.dart:

```
import 'package:flutter/material.dart';
//import 'package:lab4and5/stack_page.dart';
import 'package:lab4and5/first_page.dart';
//import "package:lab4and5/drawer_navigation_page.dart";
void main(){
  runApp(MyApp());
}

class MyApp extends StatelessWidget{
  const MyApp({super.key});


  @override
  Widget build(BuildContext context){

    return MaterialApp(debugShowCheckedModeBanner: false,
      home: //DrawerNavigationPage(),
      //StackPage()
      FirstPage(),
      );
  }
}
```

First_page.dart:

```
import 'package:flutter/material.dart';

class FirstPage extends StatelessWidget {
  const FirstPage({super.key});

  String wrapTextFromMiddle(String text) {
    int middleIndex = (text.length / 2).floor();
    return text.substring(0, middleIndex) + '\n' +
text.substring(middleIndex);
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: PreferredSize(
        preferredSize: Size.fromHeight(60.0),
        child: Container(
          decoration: BoxDecoration(
```

```
          color: Colors.teal,
          borderRadius: BorderRadius.only(
            bottomLeft: Radius.circular(15),
            bottomRight: Radius.circular(15),
          ),
        ),
        child: AppBar(
          title: const Text('This is the 3rd week homework practice'),
          titleTextStyle: TextStyle(color: Colors.white),
          centerTitle: true,
          backgroundColor: Colors.transparent,
          elevation: 0,
        ),
      ),
    ),
    backgroundColor: Colors.tealAccent,
    body: LayoutBuilder(
      builder: (context, constraints) {
        final isSmallScreen = constraints.maxWidth < 400;
        final double cardMaxWidth = isSmallScreen ? 250 : 350;
        final double cardMaxHeight = isSmallScreen ? 200 : 350;

        return Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              GestureDetector(
                onTap: () => {print('You can have the sweet recipes')},
                child: IntrinsicWidth(
                  child: IntrinsicHeight(
                    child: Card(
                      color: Colors.redAccent,
                      shape: RoundedRectangleBorder(
                        borderRadius: BorderRadius.circular(15),
                      ),
                      child: Column(
                        children: [
                          ClipRRect(
                            borderRadius: BorderRadius.only(
                              topLeft: Radius.circular(15),
                              topRight: Radius.circular(15),
                              bottomLeft: Radius.circular(5),
                              bottomRight: Radius.circular(5),
                            ),
                            child: ConstrainedBox(
                              constraints: BoxConstraints(
                                maxHeight: cardMaxHeight,
                                maxWidth: cardMaxWidth,
```

```
                        ),
                        child: Image.asset("assets/desserts.jpg"),
                      ),
                    ),
                    Container(padding: EdgeInsets.all(5)),
                    Text(
                      isSmallScreen
                          ? wrapTextFromMiddle(
                            "Please tap to browse our desserts",
                          )
                          : "Please tap to browse our desserts",
                      textAlign: TextAlign.center,
                      style: TextStyle(
                        fontWeight: FontWeight.bold,
                        fontFamily: 'RobotoMono',
                      ),
                    ),
                    Container(padding: EdgeInsets.all(5)),
                    ElevatedButton(
                      onPressed:
                          () => {print("Desserts button pressed")},
                      style: ElevatedButton.styleFrom(
                        backgroundColor: const Color.fromARGB(
                          255,
                          1,
                          61,
                          55,
                        ),
                      ),
                      child: const Text(
                        "Desserts",
                        style: TextStyle(color: Colors.white),
                      ),
                    ),
                    Container(padding: EdgeInsets.all(5)),
                  ],
                ),
              ),
            ),
          ),
        ),
        IntrinsicWidth(
          child: IntrinsicHeight(
            child: Card(
              color: Colors.redAccent,
              shape: RoundedRectangleBorder(
                borderRadius: BorderRadius.circular(15),
              ),
```

```
                    child: Column(
                      children: [
                        Stack(
                          children: [
                            ClipRRect(
                              borderRadius: BorderRadius.only(
                                topLeft: Radius.circular(15),
                                topRight: Radius.circular(15),
                                bottomLeft: Radius.circular(5),
                                bottomRight: Radius.circular(5),
                              ),
                              child: ConstrainedBox(
                                constraints: BoxConstraints(
                                  maxHeight: cardMaxHeight,
                                  maxWidth: cardMaxWidth,
                                ),
                                child: Image.asset('assets/fastfood.webp'),
                              ),
                            ),
                            Positioned(
                              bottom: 15,
                              left: 15,
                              child: Text(
                                isSmallScreen
                                    ? wrapTextFromMiddle(
                                      "This one has no button. Nah I am just
lying",

                                    )
                                    : "This one has no button. Nah I am just
lying",

                                textAlign: TextAlign.center,
                                style: TextStyle(
                                  fontWeight: FontWeight.bold,
                                  fontFamily: 'RobotoMono',
                                  color: Colors.white,
                                ),
                              ),
                            ),
                          ],
                        ),

                        //Container(padding: EdgeInsets.all(5)),
                        Container(padding: EdgeInsets.all(5)),
                        ElevatedButton(
                          style: ElevatedButton.styleFrom(
                            backgroundColor: const Color.fromARGB(
                              255,
                              1,
```

```
                                61,
                                55,
                              ),
                            ),
                            onPressed:
                                () => {print("This button was pressed fr
fr")},
                            child: Text(
                              'Fast Food',
                              style: TextStyle(color: Colors.white),
                            ),
                          ),
                          Container(padding: EdgeInsets.all(5)),
                        ],
                      ),
                    ),
                  ),
                ),
              ],
            ),
          );
        },
      ),
    );
  }
}
```

# Mobile Application Development Lab

# CSL 341

# Lab Journal 5

## Teacher: Muhammad Athesham Noor




Department of Computer Science

BAHRIA UNIVERSITY, ISLAMABAD

**Question:**

Make a flutter app that contains one stack container an image and a button centred on that image with another stack below it that displays multiple boxes in a stack placed diagonally.

Main.dart:

```
import 'package:flutter/material.dart';
import 'package:lab4and5/stack_page.dart';
//import "package:lab4and5/drawer_navigation_page.dart";
void main(){
  runApp(MyApp());
}

class MyApp extends StatelessWidget{
  const MyApp({super.key});


  @override
  Widget build(BuildContext context){

    return MaterialApp(debugShowCheckedModeBanner: false,
      home: //DrawerNavigationPage(),
      StackPage()
      );
  }
}
```

Stack_page.dart:

```
import "package:flutter/material.dart";
import "package:google_fonts/google_fonts.dart";

class StackPage extends StatelessWidget {
  StackPage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(),
      body: SizedBox(
        width: double.infinity,
        height: double.infinity,
        child: ListView(
          children: [
            Padding(
              padding: const EdgeInsets.all(8.0),
```

```
            child: Column(
              //physics: NeverScrollableScrollPhysics(),
              children: [
                Stack(
                  children: [
                    Image.asset("assets/desserts.jpg",
                        width: 400, height: 270),
                    Positioned(
                      top: 30,
                      left: 100,
                      child: Container(
                        width: 200,
                        height: 200,
                        color: Colors.amber.shade300,
                      ),
                    ),
                    Positioned(
                      top: 100,
                      left: 130,
                      child: ElevatedButton(
                        onPressed: () => print("no"),
                        child: Text("Stack Button",
                            style: GoogleFonts.nunito(
                                fontWeight: FontWeight.bold)),
                      ),
                    ),
                  ],
                ),
                Padding(
                  padding: EdgeInsets.all(10.0),
                ),
                SizedBox(
                  width: 800,
                  height: 800,
                  child:
                      ListView(scrollDirection: Axis.horizontal, children: [
                    Stack(
                      children: [
                        SizedBox(
                          width: 600,
                          height: 600,
                        ),
                        Container(
                          color: Colors.amber,
                          width: 400,
                          height: 400,
                        ),
                        Positioned(
```

```
                        top: 200,
                        left: 200,
                        child: Container(
                          color: Colors.red,
                          width: 300,
                          height: 300,
                        ),
                      ),
                      Positioned(
                        top: 350,
                        left: 350,
                        child: Container(
                          color: Colors.blue,
                          width: 200,
                          height: 200,
                        ),
                      ),
                      Positioned(
                        top: 500,
                        left: 500,
                        child: Container(
                          color: Colors.green,
                          width: 100,
                          height: 100,
                        ),
                      ),
                    ],
                  ),
                  Padding(
                    padding: EdgeInsets.all(200),
                  ),
                ]),
              ),
            ],
          ),
        ),
      ],
    ),
  ),
);
  }
}
```

# Mobile Application Development Lab

# CSL 341

# Lab Journal 06

## Teacher: Muhammad Athesham Noor




Department of Computer Science

BAHRIA UNIVERSITY, ISLAMABAD

**Question:**

Make a flutter app having multiple pages that contain a simple frontend for login, register and dashboard features.

**Main.dart:**

```dart
import "package:flutter/material.dart";
//import "package:lab6/quizpractice.dart";
import "package:lab6/main_page.dart";
import "package:lab6/dashboard_page.dart";
import "package:lab6/login_page.dart";
import "package:lab6/register_page.dart";


void main(){
  runApp(MyApp());
}

class MyApp extends StatelessWidget{
  const MyApp({super.key});

  @override
  Widget build(BuildContext context){
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: "Quiz Practice App",
      initialRoute: "/",
      routes: {
        "/": (context) => MainPage(),
        "/login": (context) => LoginPage(),
        "/register": (context) => RegisterPage(),
        "/dashboard": (context) => DashboardPage(),
      },
    );
  }
}
```

Dashboard_page:

```dart
//LAB 6 Do Not Delete
import "package:flutter/material.dart";

class DashboardPage extends StatelessWidget {
  const DashboardPage({super.key});

  @override
  Widget build(BuildContext context) {
```

```
    return Scaffold(
      backgroundColor: Colors.tealAccent,
      appBar: AppBar(
        backgroundColor: Colors.teal,
        title: const Text("This is the messaging app"),
        centerTitle: true,
        titleTextStyle: const TextStyle(
            fontWeight: FontWeight.bold,
            fontFamily: "MonoRobotto",
            color: Colors.black,
            fontSize: 12.0),
      ),
      body: ListView(
        //scrollDirection: Axis.horizontal,
        padding: const EdgeInsets.all(20.0),
        children: [
          CustomScrollWidget(
            name: Text("Want Desserts?", style: TextStyle(fontWeight:
FontWeight.bold, fontFamily: "MonoRobotto", fontSize: 15.0),),
            current: Image.asset("assets/desserts.jpg"),
          ),
          SizedBox(height: 20),
          CustomScrollWidget(
            name: Text("or want Fast Food?", style: TextStyle(fontWeight:
FontWeight.bold, fontFamily: "MonoRobotto", fontSize: 15.0),),
            current: Image.asset("assets/fastfood.webp"),
          ),
        ],
      ),
    );
  }
}

class CustomScrollWidget extends StatelessWidget {
  final Text name;
  final Image current;
  const CustomScrollWidget({
    super.key,
    required this.name,
    required this.current,
  });

  @override
  Widget build(BuildContext context) {
    return Center(
      child: IntrinsicWidth(
          child: IntrinsicHeight(
            child: Card(
```

```
              color: const Color.fromARGB(255, 73, 178, 127),
              shape: RoundedRectangleBorder(
                borderRadius: BorderRadius.only(
                  topLeft: Radius.circular(50),
                  topRight: Radius.circular(15),
                  bottomLeft: Radius.circular(15),
                  bottomRight: Radius.circular(15),
                ),
              ),
              elevation: 5.0,
              shadowColor: Colors.tealAccent,
              child: Column(
                  mainAxisAlignment: MainAxisAlignment.start,
                  children: [
                    FittedBox(
                      fit: BoxFit.contain,
                      child: ClipRRect(
                        borderRadius: BorderRadius.only(
                          topLeft: Radius.circular(50),
                          topRight: Radius.circular(15),
                        ),
                        child: Container(
                          constraints: BoxConstraints(
                            maxWidth: 200.0,
                            maxHeight: 200.0,
                          ),
                          child: current,
                        ),
                      ),
                    ),
                    name,
                  ],
                ),
            ),
          ),
        ),
    );
  }
}
```

Login_page:

```
//LAB 6 Do Not Delete


import "package:flutter/material.dart";
```

```dart
class LoginPage extends StatelessWidget {
  const LoginPage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: Colors.tealAccent,
      appBar: AppBar(
        backgroundColor: Colors.teal,
        title: const Text("This is the messaging app"),
        centerTitle: true,
        titleTextStyle: const TextStyle(
            fontWeight: FontWeight.bold,
            fontFamily: "MonoRobotto",
            color: Colors.black,
            fontSize: 12.0),
      ),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            SizedBox(
              width: 200,
              height: 100,
              child: TextField(
                decoration: InputDecoration(
                  fillColor: Colors.white,
                  filled: true,
                  labelText: "Username",
                  border: OutlineInputBorder(
                    borderRadius: BorderRadius.circular(15),
                    borderSide: BorderSide(
                      color: Colors.black,
                      width: 3,
                      style: BorderStyle.solid,
                    ),
                  ),
                ),
              ),
            ),
            SizedBox(
              width: 200,
              height: 100,
              child: TextField(
                decoration: InputDecoration(
                  fillColor: Colors.white,
                  filled: true,
                  labelText: "Password",
```

```
                  border: OutlineInputBorder(
                    borderRadius: BorderRadius.circular(15),
                    borderSide: BorderSide(
                      color: Colors.black,
                      width: 3,
                      style: BorderStyle.solid,
                    ),
                  ),
                ),
              ),
            ),
            ElevatedButton(
              onPressed: () => Navigator.pushNamed(context, "/dashboard"),
              child: const Text("Login"),
            ),
            Container(
                padding: EdgeInsets.all(20),
              ),
            ElevatedButton(
              onPressed: () => Navigator.pushNamed(context, "/register"),
              child: const Text("Register"),
            ),
          ],
        ),
      ),
    );
  }
}
```

Main_page.dart:

```
//LAB 6 Do Not Delete
import "package:flutter/material.dart";

class MainPage extends StatelessWidget {
  const MainPage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: Colors.tealAccent,
      appBar: AppBar(
        backgroundColor: Colors.teal,
        title: const Text("This is the messaging app"),
        centerTitle: true,
        titleTextStyle: const TextStyle(
            fontWeight: FontWeight.bold,
```

```
            fontFamily: "MonoRobotto",
            color: Colors.black,
            fontSize: 12.0),
      ),
      body: Center(
        child: Container(
          decoration: BoxDecoration(
            color: Colors.white,
            borderRadius: BorderRadius.circular(15),
          ),
          constraints: BoxConstraints(
            maxHeight: 200.0,
            maxWidth: 200.0,
          ),
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: [
              ElevatedButton(
                  onPressed: () => Navigator.pushNamed(context, "/register"),
                  child: const Text("Register")),
              Container(
                padding: EdgeInsets.all(20),
              ),
              ElevatedButton(
                  onPressed: () => Navigator.pushNamed(context, "/login"),
                  child: const Text("Login")),
            ],
          ),
        ),
      ),
    );
  }
}
```

Register_page.dart:

```
//LAB 6 Do Not Delete

import "package:flutter/material.dart";

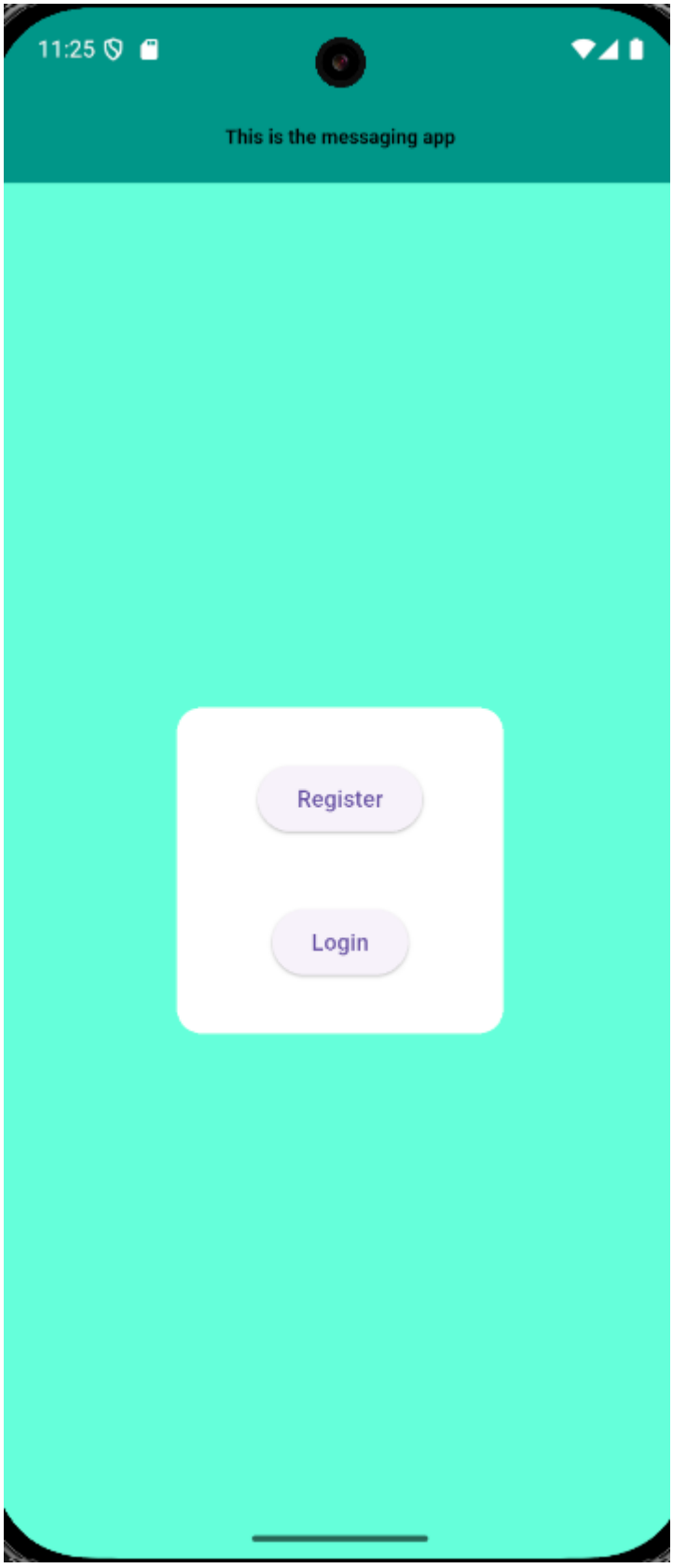
class RegisterPage extends StatelessWidget {
  const RegisterPage({super.key});

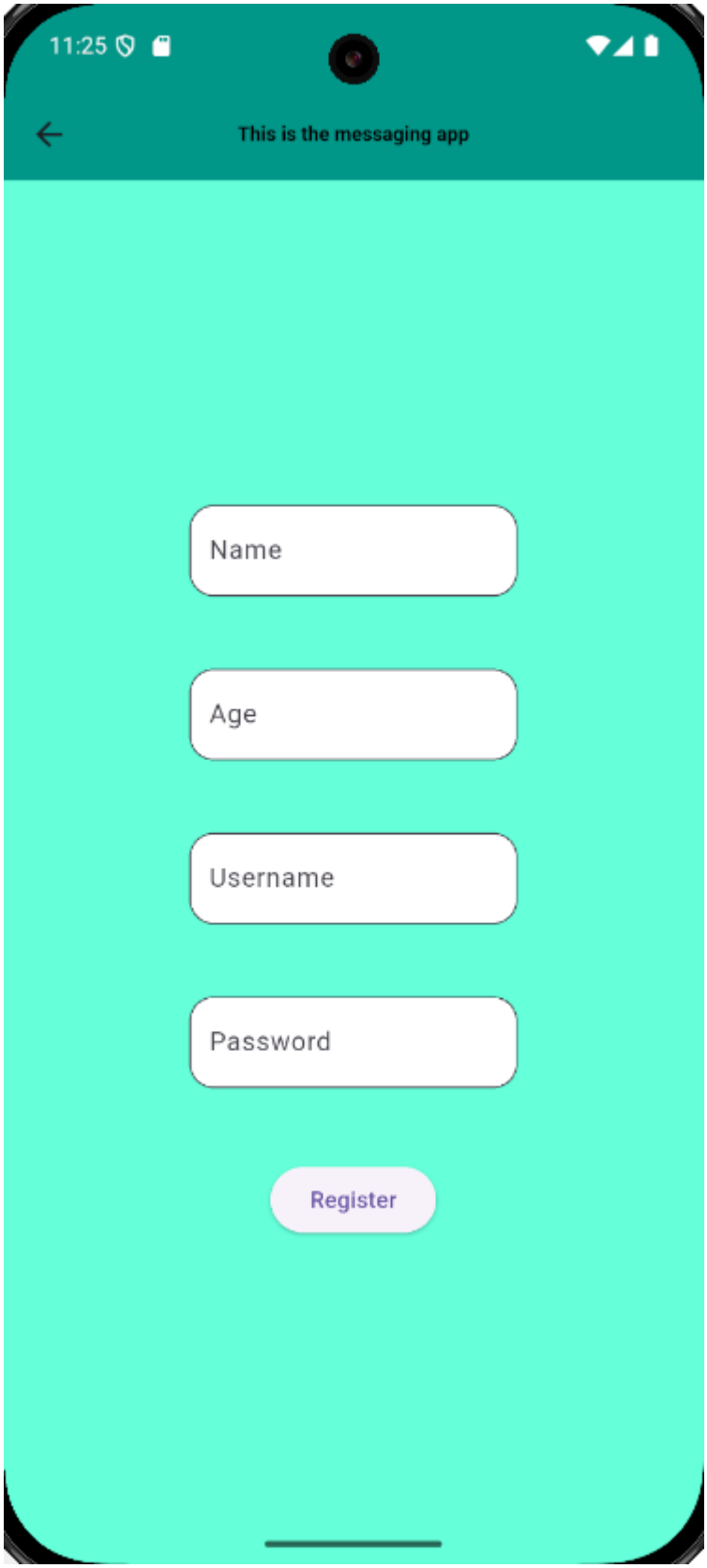
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: Colors.tealAccent,
```

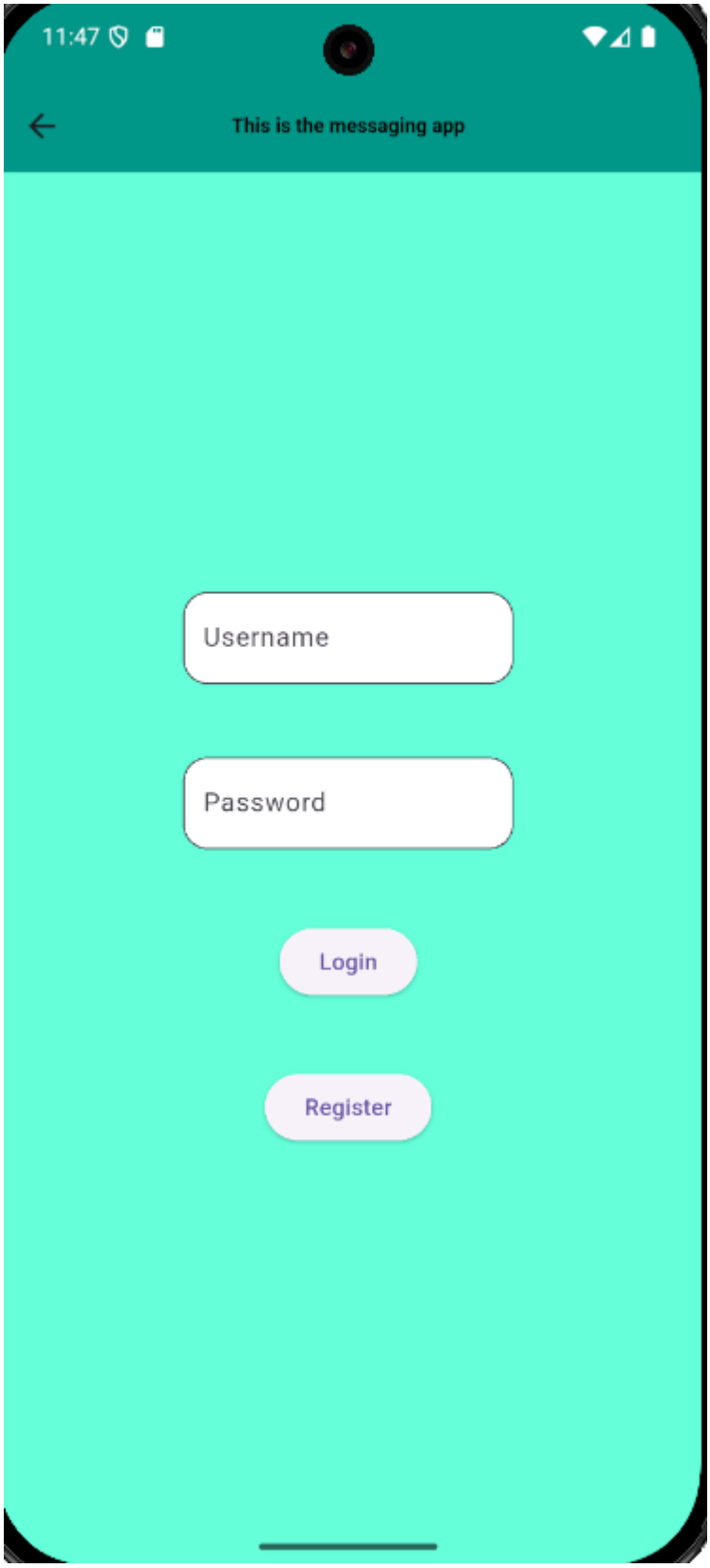
```
      appBar: AppBar(
        backgroundColor: Colors.teal,
        title: const Text("This is the messaging app"),
        centerTitle: true,
        titleTextStyle: const TextStyle(
            fontWeight: FontWeight.bold,
            fontFamily: "MonoRobotto",
            color: Colors.black,
            fontSize: 12.0),
      ),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            SizedBox(
              width: 200,
              height: 100,
              child: TextField(
                decoration: InputDecoration(
                  fillColor: Colors.white,
                  filled: true,
                  labelText: "Name",
                  border: OutlineInputBorder(
                    borderRadius: BorderRadius.circular(15),
                    borderSide: BorderSide(
                      color: Colors.black,
                      width: 3,
                      style: BorderStyle.solid,
                    ),
                  ),
                ),
              ),
            ),
            SizedBox(
              width: 200,
              height: 100,
              child: TextField(
                decoration: InputDecoration(
                  fillColor: Colors.white,
                  filled: true,
                  labelText: "Age",
                  border: OutlineInputBorder(
                    borderRadius: BorderRadius.circular(15),
                    borderSide: BorderSide(
                      color: Colors.black,
                      width: 3,
                      style: BorderStyle.solid,
                    ),
```

```
              ),
            ),
          ),
        ),
        SizedBox(
          width: 200,
          height: 100,
          child: TextField(
            decoration: InputDecoration(
              fillColor: Colors.white,
              filled: true,
              labelText: "Username",
              border: OutlineInputBorder(
                borderRadius: BorderRadius.circular(15),
                borderSide: BorderSide(
                  color: Colors.black,
                  width: 3,
                  style: BorderStyle.solid,
                ),
              ),
            ),
          ),
        ),
        SizedBox(
          width: 200,
          height: 100,
          child: TextField(
            decoration: InputDecoration(
              fillColor: Colors.white,
              filled: true,
              labelText: "Password",
              border: OutlineInputBorder(
                borderRadius: BorderRadius.circular(15),
                borderSide: BorderSide(
                  color: Colors.black,
                  width: 3,
                  style: BorderStyle.solid,
                ),
              ),
            ),
          ),
        ),
        ElevatedButton(
          onPressed: () => Navigator.pushNamed(context, "/dashboard"),
          child: const Text("Register"),
        ),
      ],
    ),
```

```
      ),
    );
  }
}
```


11:25
This is the messaging app
Register
Login


11:25
This is the messaging app
Name
Age
Username
Password
Register


11:47
This is the messaging app
Username
Password
Login
Register

# Mobile Application Development Lab

# CSL 341

# Lab Journal 7

## Teacher: Muhammad Athesham Noor




Department of Computer Science

BAHRIA UNIVERSITY, ISLAMABAD

**Question:**

Make a flutter app that has a drawer, Floating action button, ListView with multiple items and a bottom navigation bar.

Main.dart:

```
import 'package:flutter/material.dart';
//import 'package:lab4and5/stack_page.dart';
import "package:lab4and5/drawer_navigation_page.dart";
void main(){
  runApp(MyApp());
}

class MyApp extends StatelessWidget{
  const MyApp({super.key});


  @override
  Widget build(BuildContext context){

    return MaterialApp(debugShowCheckedModeBanner: false,
      home: DrawerNavigationPage(),
      //StackPage()
      );
  }
}
```

drawer_navigation_page.dart:

```
import "package:flutter/material.dart";
import "package:google_fonts/google_fonts.dart";
import "package:lab4and5/stack_page.dart";
import "package:lab4and5/first_page.dart";

class DrawerNavigationPage extends StatefulWidget{
  DrawerNavigationPage({super.key});

  DrawerNavigationPageState createState() => DrawerNavigationPageState();
}

class DrawerNavigationPageState extends State<DrawerNavigationPage> {
  var currArray = [
    "Item 1",
    "Item 2",
    "Item 3",
    "Item 4",
    "Item 5",
```

```
    "Item 6"
  ];
  int currIndex = 0;

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text("This is the Drawer and Navigaton page", style:
GoogleFonts.nunito()),
        centerTitle: true,
        backgroundColor: Colors.blue,
      ),
      drawer: Drawer(
        child: Container(
          color: Colors.blueGrey,
          child: ListView(
            children:[
              DrawerHeader(
                child: Image.asset("assets/desserts.jpg")
              ),
              ListTile(
                leading: Icon(Icons.home),
                title: Text("Item 1"),
                onTap: () => Navigator.push(context,
MaterialPageRoute(builder: (context) => FirstPage())),
                trailing: Icon(Icons.dangerous),
                ),
              ListTile(
                leading: Icon(Icons.visibility),
                trailing: Icon(Icons.dangerous),
                title: Text("Item 2"),
                onTap: () => Navigator.push(context,
MaterialPageRoute(builder: (context) => StackPage()))
                ),
            ],
          ),
        ),
      ),
      body: Center(
        child: ListView.builder(
          physics: NeverScrollableScrollPhysics(),
          itemCount: currArray.length,
          itemBuilder: (context, index){
            return ListTile(
              leading: Icon(Icons.visibility),
              title: Text(currArray[index], style: GoogleFonts.nunito()),
              trailing: Icon(Icons.dangerous),
```

```
            );
          }
        ),
      ),

      bottomNavigationBar: BottomNavigationBar(
        currentIndex: currIndex,
        onTap: (int index){
          setState(() => currIndex = index);
        },
        items: [
          BottomNavigationBarItem(
            icon: Icon(Icons.add),
            label: "Item 1",
            ),
            BottomNavigationBarItem(
            icon: Icon(Icons.remove),
            label: "Item 2",
            )
        ],
        ),
        floatingActionButton: FloatingActionButton(onPressed: () =>
print("no"), child: Icon(Icons.add)),
        floatingActionButtonLocation:
FloatingActionButtonLocation.centerFloat,
    );
  }
}
```

Stack_page.dart:

```
import "package:flutter/material.dart";
import "package:google_fonts/google_fonts.dart";

class StackPage extends StatelessWidget {
  StackPage({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(),
      body: SizedBox(
        width: double.infinity,
        height: double.infinity,
        child: ListView(
          children: [
            Padding(
              padding: const EdgeInsets.all(8.0),
```

```
            child: Column(
              //physics: NeverScrollableScrollPhysics(),
              children: [
                Stack(
                  children: [
                    Image.asset("assets/desserts.jpg",
                        width: 400, height: 270),
                    Positioned(
                      top: 30,
                      left: 100,
                      child: Container(
                        width: 200,
                        height: 200,
                        color: Colors.amber.shade300,
                      ),
                    ),
                    Positioned(
                      top: 100,
                      left: 130,
                      child: ElevatedButton(
                        onPressed: () => print("no"),
                        child: Text("Stack Button",
                            style: GoogleFonts.nunito(
                                fontWeight: FontWeight.bold)),
                      ),
                    ),
                  ],
                ),
                Padding(
                  padding: EdgeInsets.all(10.0),
                ),
                SizedBox(
                  width: 800,
                  height: 800,
                  child:
                      ListView(scrollDirection: Axis.horizontal, children: [
                    Stack(
                      children: [
                        SizedBox(
                          width: 600,
                          height: 600,
                        ),
                        Container(
                          color: Colors.amber,
                          width: 400,
                          height: 400,
                        ),
                        Positioned(
```

```
                            top: 200,
                            left: 200,
                            child: Container(
                              color: Colors.red,
                              width: 300,
                              height: 300,
                            ),
                          ),
                          Positioned(
                            top: 350,
                            left: 350,
                            child: Container(
                              color: Colors.blue,
                              width: 200,
                              height: 200,
                            ),
                          ),
                          Positioned(
                            top: 500,
                            left: 500,
                            child: Container(
                              color: Colors.green,
                              width: 100,
                              height: 100,
                            ),
                          ),
                        ],
                      ),
                      Padding(
                        padding: EdgeInsets.all(200),
                      ),
                    ]),
                  ),
                ],
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

First_page.dart:

```
import 'package:flutter/material.dart';

class FirstPage extends StatelessWidget {
  const FirstPage({super.key});
```

```
  String wrapTextFromMiddle(String text) {
    int middleIndex = (text.length / 2).floor();
    return text.substring(0, middleIndex) + '\n' +
text.substring(middleIndex);
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: PreferredSize(
        preferredSize: Size.fromHeight(60.0),
        child: Container(
          decoration: BoxDecoration(
            color: Colors.teal,
            borderRadius: BorderRadius.only(
              bottomLeft: Radius.circular(15),
              bottomRight: Radius.circular(15),
            ),
          ),
          child: AppBar(
            title: const Text('This is the 3rd week homework practice'),
            titleTextStyle: TextStyle(color: Colors.white),
            centerTitle: true,
            backgroundColor: Colors.transparent,
            elevation: 0,
          ),
        ),
      ),
      backgroundColor: Colors.tealAccent,
      body: LayoutBuilder(
        builder: (context, constraints) {
          final isSmallScreen = constraints.maxWidth < 400;
          final double cardMaxWidth = isSmallScreen ? 250 : 350;
          final double cardMaxHeight = isSmallScreen ? 200 : 350;

          return Center(
            child: Column(
              mainAxisAlignment: MainAxisAlignment.center,
              children: [
                GestureDetector(
                  onTap: () => {print('You can have the sweet recipes')},
                  child: IntrinsicWidth(
                    child: IntrinsicHeight(
                      child: Card(
                        color: Colors.redAccent,
                        shape: RoundedRectangleBorder(
                          borderRadius: BorderRadius.circular(15),
```

```
                    ),
                    child: Column(
                      children: [
                        ClipRRect(
                          borderRadius: BorderRadius.only(
                            topLeft: Radius.circular(15),
                            topRight: Radius.circular(15),
                            bottomLeft: Radius.circular(5),
                            bottomRight: Radius.circular(5),
                          ),
                          child: ConstrainedBox(
                            constraints: BoxConstraints(
                              maxHeight: cardMaxHeight,
                              maxWidth: cardMaxWidth,
                            ),
                            child: Image.asset("assets/desserts.jpg"),
                          ),
                        ),
                        Container(padding: EdgeInsets.all(5)),
                        Text(
                          isSmallScreen
                              ? wrapTextFromMiddle(
                                "Please tap to browse our desserts",
                              )
                              : "Please tap to browse our desserts",
                          textAlign: TextAlign.center,
                          style: TextStyle(
                            fontWeight: FontWeight.bold,
                            fontFamily: 'RobotoMono',
                          ),
                        ),
                        Container(padding: EdgeInsets.all(5)),
                        ElevatedButton(
                          onPressed:
                              () => {print("Desserts button pressed")},
                          style: ElevatedButton.styleFrom(
                            backgroundColor: const Color.fromARGB(
                              255,
                              1,
                              61,
                              55,
                            ),
                          ),
                          child: const Text(
                            "Desserts",
                            style: TextStyle(color: Colors.white),
                          ),
                        ),
```

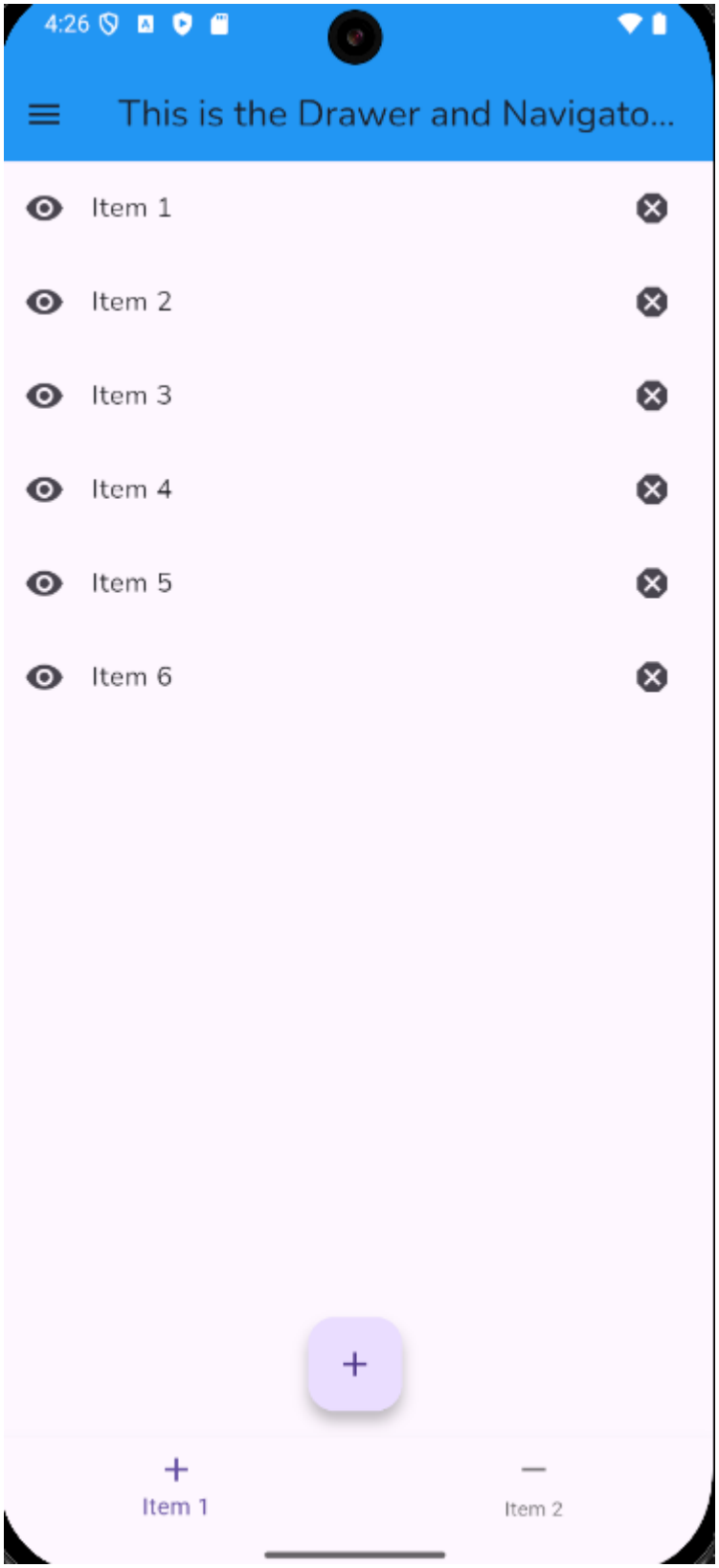
```
                        Container(padding: EdgeInsets.all(5)),
                      ],
                    ),
                  ),
                ),
              ),
            ),
            IntrinsicWidth(
              child: IntrinsicHeight(
                child: Card(
                  color: Colors.redAccent,
                  shape: RoundedRectangleBorder(
                    borderRadius: BorderRadius.circular(15),
                  ),
                  child: Column(
                    children: [
                      Stack(
                        children: [
                          ClipRRect(
                            borderRadius: BorderRadius.only(
                              topLeft: Radius.circular(15),
                              topRight: Radius.circular(15),
                              bottomLeft: Radius.circular(5),
                              bottomRight: Radius.circular(5),
                            ),
                            child: ConstrainedBox(
                              constraints: BoxConstraints(
                                maxHeight: cardMaxHeight,
                                maxWidth: cardMaxWidth,
                              ),
                              child: Image.asset('assets/fastfood.webp'),
                            ),
                          ),
                          Positioned(
                            bottom: 15,
                            left: 15,
                            child: Text(
                              isSmallScreen
                                  ? wrapTextFromMiddle(
                                    "This one has no button. Nah I am just
lying",

                                  )
                                  : "This one has no button. Nah I am just
lying",
                              textAlign: TextAlign.center,
                              style: TextStyle(
                                fontWeight: FontWeight.bold,
                                fontFamily: 'RobotoMono',
```

```
                                color: Colors.white,
                              ),
                            ),
                          ),
                        ],
                      ),

                      //Container(padding: EdgeInsets.all(5)),
                      Container(padding: EdgeInsets.all(5)),
                      ElevatedButton(
                        style: ElevatedButton.styleFrom(
                          backgroundColor: const Color.fromARGB(
                            255,
                            1,
                            61,
                            55,
                          ),
                        ),
                        onPressed:
                            () => {print("This button was pressed fr
fr")},

                        child: Text(
                          'Fast Food',
                          style: TextStyle(color: Colors.white),
                        ),
                      ),
                      Container(padding: EdgeInsets.all(5)),
                    ],
                  ),
                ),
              ),
            ),
          ],
        ),
      );
    },
  ),
 );
 }
}
```


4:26
This is the Drawer and Navigato...
Item 1
Item 2
Item 3
Item 4
Item 5
Item 6
Item 1
Item 2

# Mobile Application Development Lab

# CSL 341

# Lab Journal 09

## Teacher: Muhammad Athesham Noor



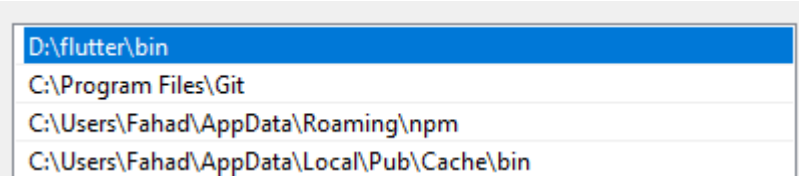

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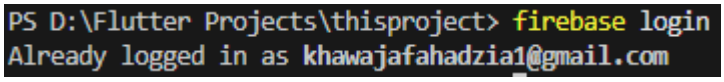
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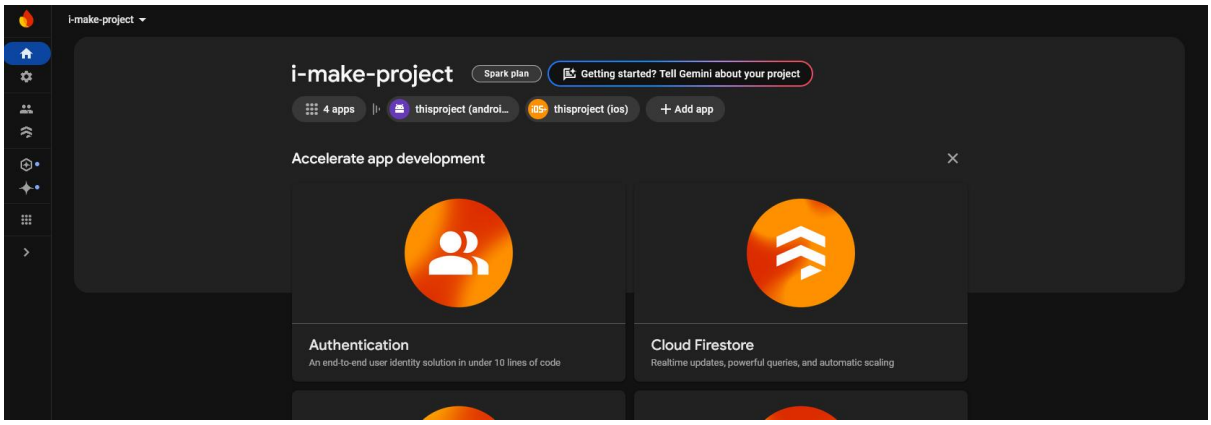
**Question:**

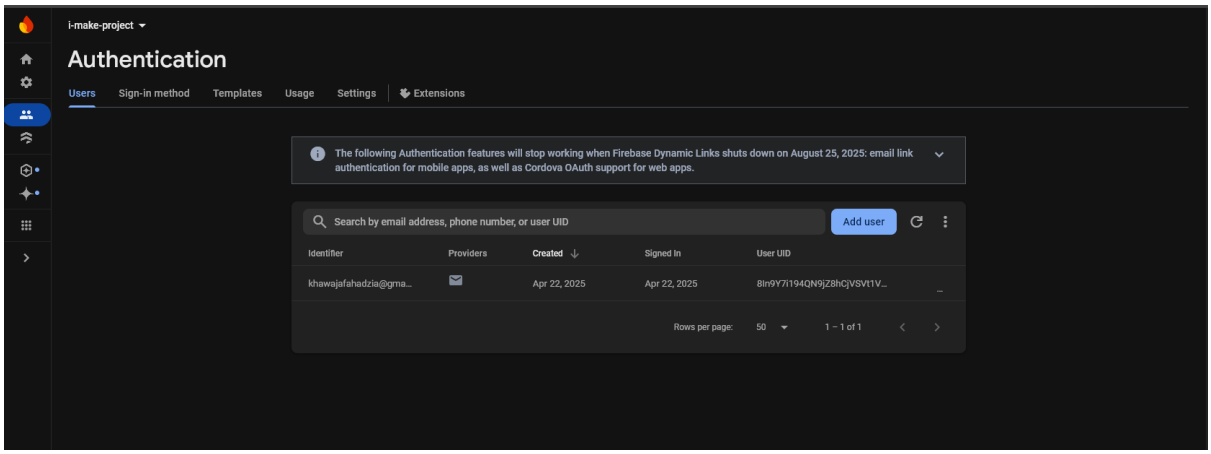
Set up Firebase authentication in a flutter app.

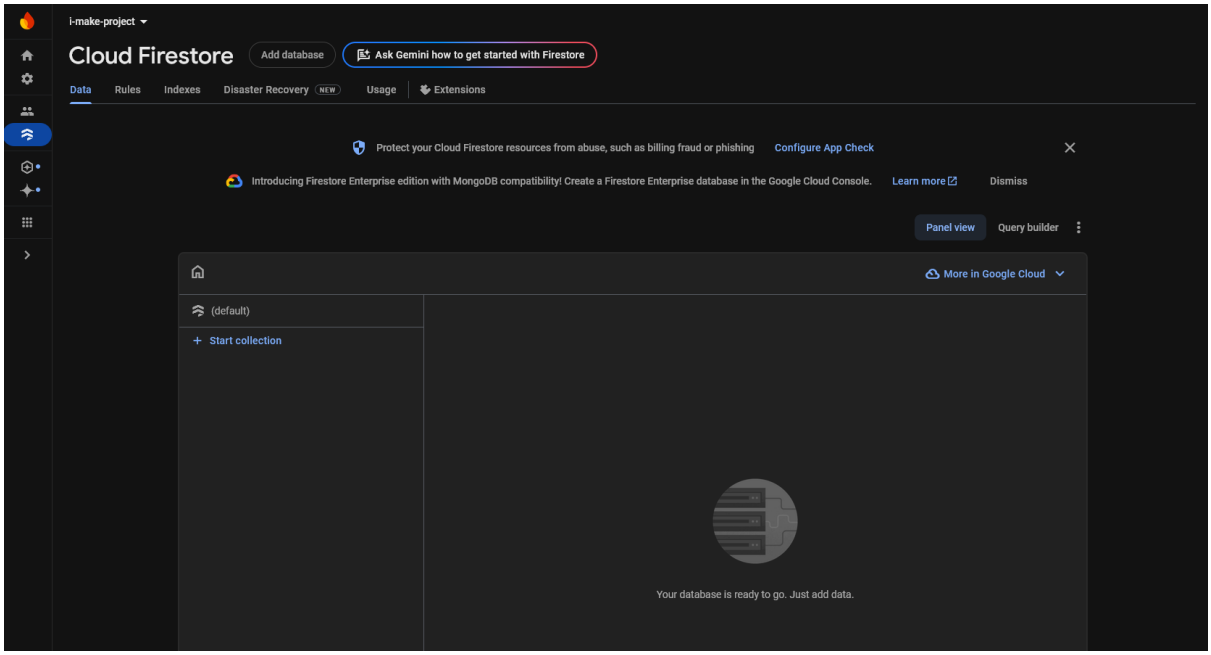
**Connection:**

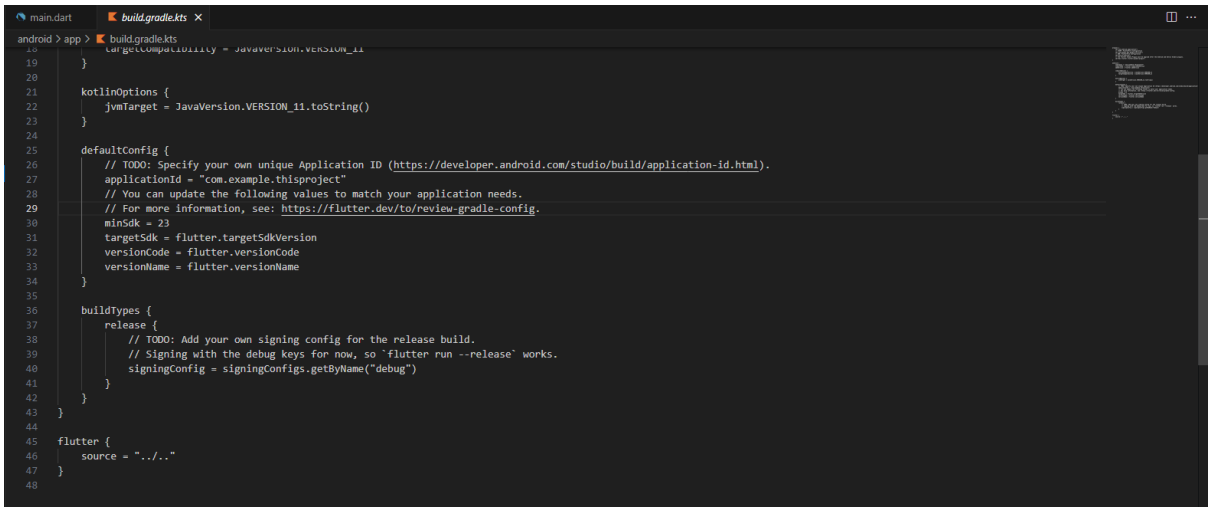


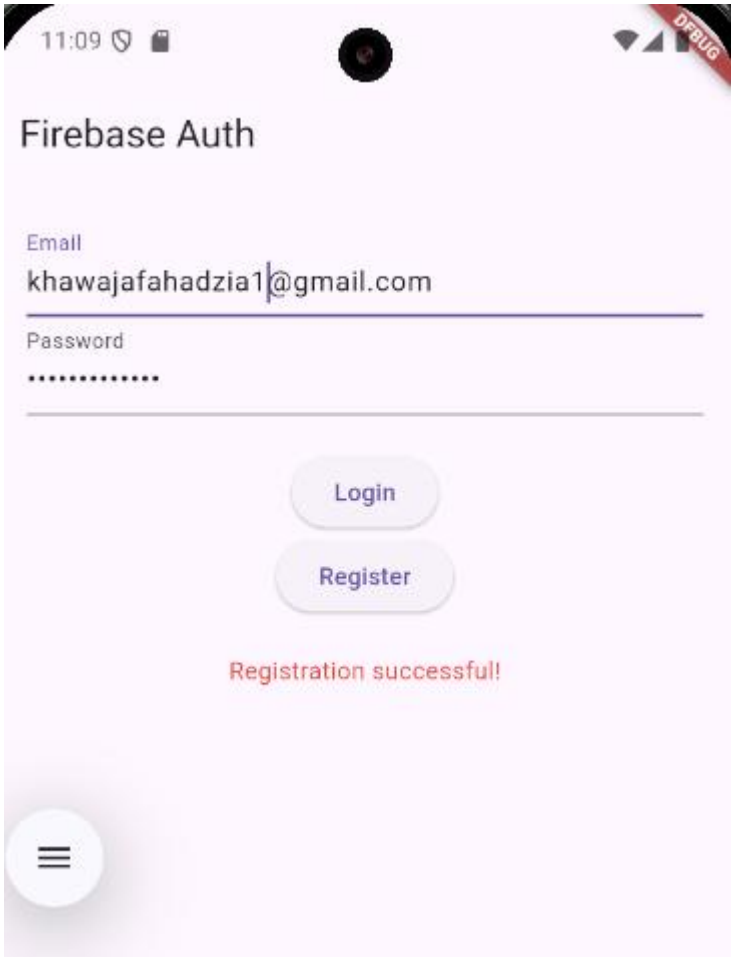



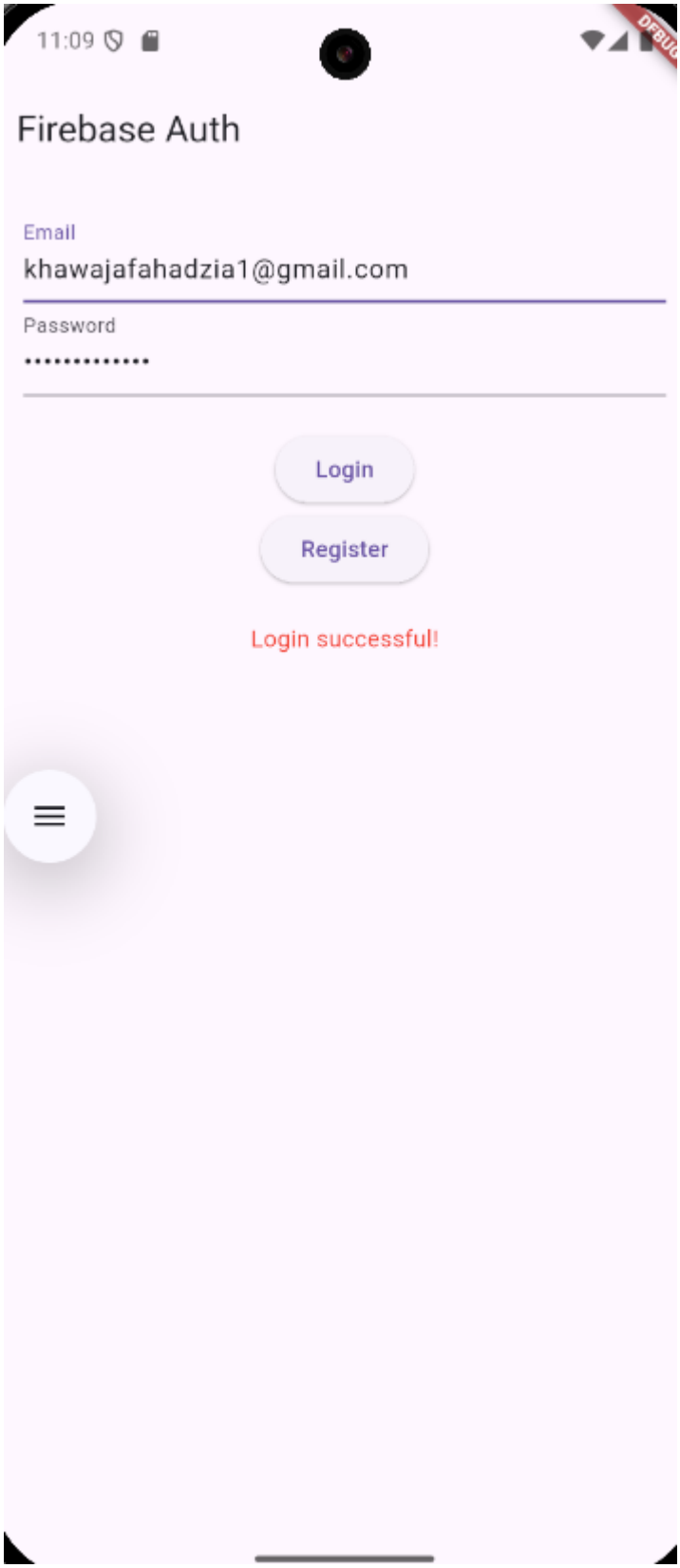




i-make-project
Cloud Firestore
Add database
Ask Gemini how to get started with Firestore
Data
Rules
Indexes
Disaster Recovery
NEW
Usage
Extensions
Protect your Cloud Firestore resources from abuse, such as billing fraud or phishing
Configure App Check
Introducing Firestore Enterprise edition with MongoDB compatibility! Create a Firestore Enterprise database in the Google Cloud Console.
Learn more
Dismiss
Panel view
Query builder
More in Google Cloud
(default)
Start collection
Your database is ready to go. Just add data.


main.dart
build.gradle.kts
android > app > build.gradle.kts
}
kotlinOptions {
jvmTarget = JavaVersion.VERSION_11.toString()
}
defaultConfig {
// TODO: Specify your own unique Application ID (https://developer.android.com/studio/build/application-id.html).
applicationId = "com.example.thisproject"
// You can update the following values to match your application needs.
// For more information, see: https://flutter.dev/to/review-gradle-config.
minSdk = 23
targetSdk = flutter.targetSdkVersion
versionCode = flutter.versionCode
versionName = flutter.versionName
}
buildTypes {
release {
// TODO: Add your own signing config for the release build.
// Signing with the debug keys for now, so `flutter run --release` works.
signingConfig = signingConfigs.getByName("debug")
}
}
}
flutter {
source = "../.."
}


11:09
DEBUG
Firebase Auth
Email
khawajafahadzia1@gmail.com
Password
Login
Register
Registration successful!




Main.dart:

```
import 'package:flutter/material.dart';
```

```dart
import 'package:firebase_core/firebase_core.dart';
import 'package:firebase_auth/firebase_auth.dart';

void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp();
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(title: 'Firebase Auth Demo', home: AuthScreen());
  }
}

class AuthScreen extends StatefulWidget {
  @override
  _AuthScreenState createState() => _AuthScreenState();
}

class _AuthScreenState extends State<AuthScreen> {
  final FirebaseAuth _auth = FirebaseAuth.instance;
  final emailController = TextEditingController();
  final passwordController = TextEditingController();
  String message = '';
  Future<void> signIn() async {
    try {
      await _auth.signInWithEmailAndPassword(
        email: emailController.text.trim(),
        password: passwordController.text.trim(),
      );
      setState(() => message = 'Login successful!');
    } catch (e) {
      setState(() => message = 'Login failed: $e');
    }
  }

  Future<void> register() async {
    try {
      await _auth.createUserWithEmailAndPassword(
        email: emailController.text.trim(),
        password: passwordController.text.trim(),
      );
      setState(() => message = 'Registration successful!');
    } catch (e) {
      setState(() => message = 'Registration failed: $e');
    }
```

```
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Firebase Auth')),
      body: Padding(
        padding: EdgeInsets.all(20),
        child: Column(
          children: [
            TextField(
              controller: emailController,
              decoration: InputDecoration(labelText: 'Email'),
            ),
            TextField(
              controller: passwordController,
              decoration: InputDecoration(labelText: 'Password'),
              obscureText: true,
            ),
            const SizedBox(height: 20),
            ElevatedButton(onPressed: signIn, child: Text('Login')),
            ElevatedButton(onPressed: register, child: Text('Register')),
            const SizedBox(height: 20),
            Text(message, style: TextStyle(color: Colors.red)),
          ],
        ),
      ),
    );
  }
}
```

# Mobile Application Development Lab

# CSL 341

# Lab Journal 10

## Teacher: Muhammad Athesham Noor

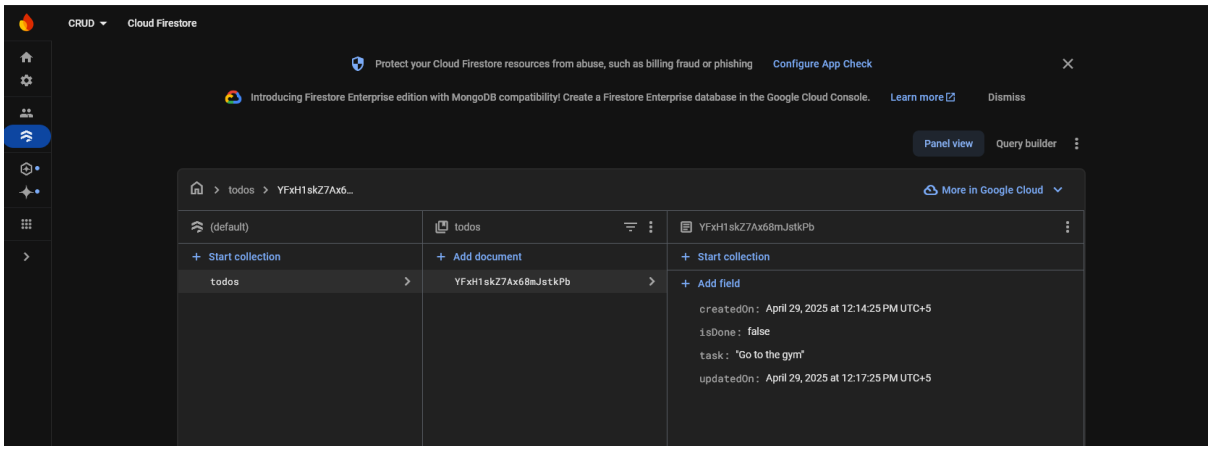


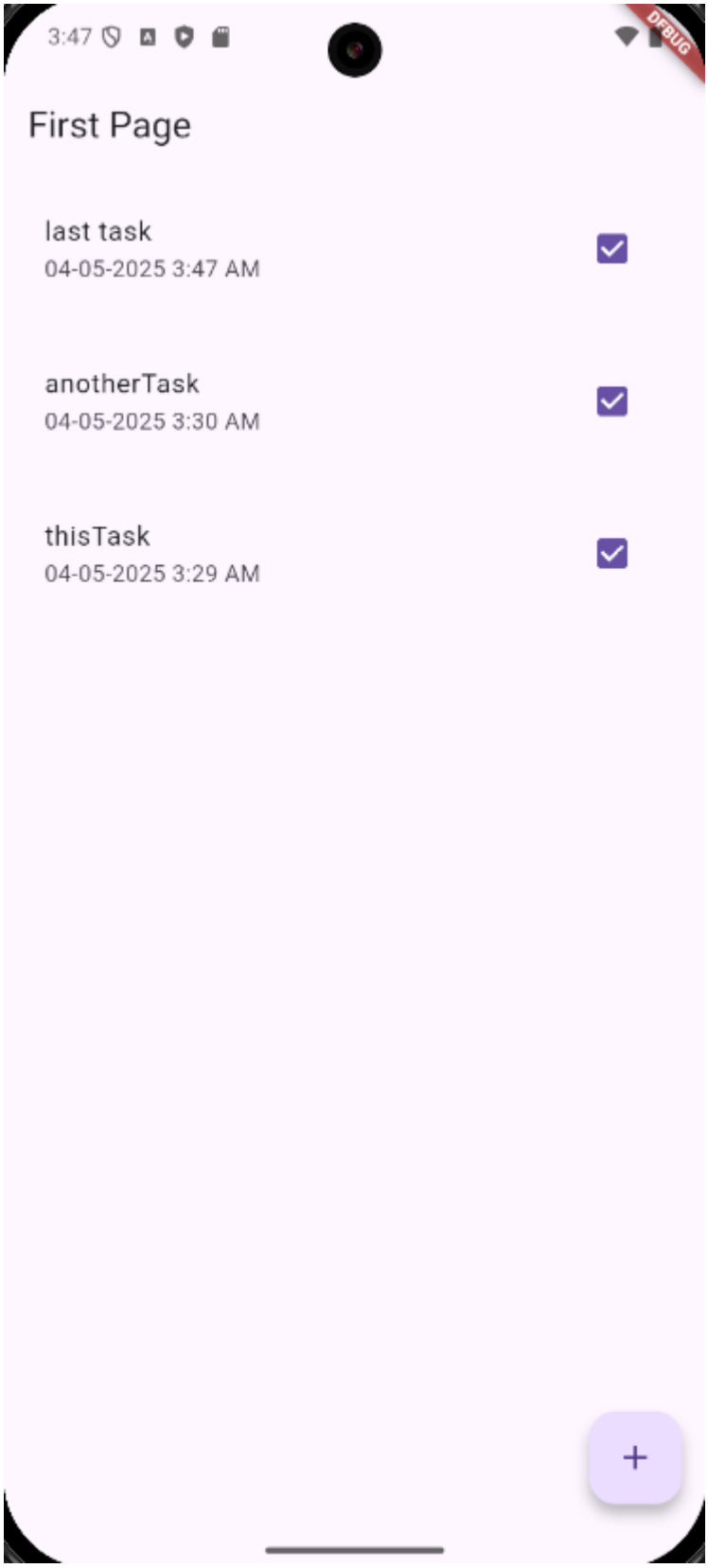

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**Question:**

Set up CRUD functionality in a flutter app using Firebase Firestore.






Main.dart:

```
import 'package:flutter/material.dart';
import 'package:firebase_core/firebase_core.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
```

```dart
import 'package:lab10/first_page.dart';

void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp();
  FirebaseFirestore.instance.settings = const Settings(
    persistenceEnabled: true,
    cacheSizeBytes: Settings.CACHE_SIZE_UNLIMITED,
  );
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter Demo',
      home: MyHomePage(),
    );
  }
}
```

Todos.dart:

```dart
import 'package:cloud_firestore/cloud_firestore.dart';

class Todos{
  String task;
  bool isDone;
  Timestamp createdOn;
  Timestamp updatedOn;
  Todos({ required this.task, required this.isDone, required this.createdOn,
required this.updatedOn});

  Todos.fromJson(Map<String, dynamic> json)
  : this( task: json['task'], isDone: json['isDone'], createdOn:
json['createdOn'], updatedOn: json['updatedOn']);

  Todos copyWith({
    String? task,
    bool? isDone,
    Timestamp? createdOn,
    Timestamp? updatedOn,
  }){
    return Todos(
      task: task ?? this.task,
```

```
      isDone: isDone ?? this.isDone,
      createdOn: createdOn ?? this.createdOn,
      updatedOn: updatedOn ?? this.updatedOn,
    );
  }

  Map<String, dynamic> toJson() {
    return {
      'task': task,
      'isDone': isDone,
      'createdOn': createdOn,
      'updatedOn': updatedOn,
    };
  }
}
```

Database_service.dart:

```
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:lab10/models/todos.dart';

const String todosCollection = 'todos';

class DatabaseService {
  final FirebaseFirestore _instance = FirebaseFirestore.instance;
  late final CollectionReference _todosReference;

  DatabaseService() {
    _todosReference = _instance
        .collection(todosCollection)
        .withConverter<Todos>(
          fromFirestore: (snapshots, _) => Todos.fromJson(snapshots.data()!),
          toFirestore: (todos, _) => todos.toJson(),
        );
  }

  Stream<QuerySnapshot> getTodos() {
    return _todosReference.snapshots();
  }

  void addTodos(Todos todos) async {
    try{
      _todosReference.add(todos);
    }
    catch(e) {
      print('Error adding todo: $e');
    }
  }

  void updateTodo(String id, Todos todo) async {
```

```
    try{
      await _todosReference.doc(id).update(todo.toJson());
    }
    catch(e) {
      print('Error updating todo: $e');
    }
  }

  void deleteTodo(String id) async{
    try{
      await _todosReference.doc(id).delete();
    }
    catch(e) {
      print('Error deleting todo: $e');
    }
  }
}
```

First_page.dart:

```
import 'package:flutter/material.dart';


import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:lab10/services/database_service.dart';
import 'package:lab10/models/todos.dart';
import 'package:intl/intl.dart';

class MyHomePage extends StatefulWidget {
  const MyHomePage({super.key});

  @override
  State<MyHomePage> createState() => _MyHomePageState();
}

class _MyHomePageState extends State<MyHomePage> {

  final DatabaseService _databaseService = DatabaseService();
  TextEditingController _controller = TextEditingController();

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('First Page'),
      ),
      floatingActionButton: FloatingActionButton(
        child: const Icon(Icons.add),
        onPressed: () async {
          return showDialog(
```

```
            context: context,
            builder: (context) {
              return AlertDialog(
                title: const Text('Add Todo'),
                content: TextField(
                  controller: _controller,
                  decoration: const InputDecoration(hintText: 'Enter task'),
                ),
                actions: [
                  TextButton(
                    onPressed: () {
                      _databaseService.addTodos(Todos(task: _controller.text,
isDone: false, createdOn: Timestamp.now(), updatedOn: Timestamp.now()));
                      Navigator.pop(context);
                      },
                    child: const Text('Add'),
                  ),
                ],
              );
            },
          );
        },
      ),
      body: SafeArea(
        child: Column(
          children: [
            SizedBox(
              height: MediaQuery.sizeOf(context).height * 0.8,
              width: MediaQuery.sizeOf(context).width,
              child: StreamBuilder(
                stream: _databaseService.getTodos(),
                builder: (context, snapshot){
                  if(snapshot.hasError){
                    return const Center(child: Text('Error'));
                  }
                  if(snapshot.connectionState == ConnectionState.waiting){
                    return const Center(child: CircularProgressIndicator());
                  }
                  List todos = snapshot.data?.docs ?? [];
                  if(todos.isEmpty){
                    return const Center(child: Text('Add a todo!'));
                  }
                  return ListView.builder(
                    itemCount: todos.length,
                    itemBuilder: (context, index){
                      final Todos todo = todos[index].data();
                      final String id = todos[index].id;
                      return Padding(
```

```
                        padding: const EdgeInsets.all(10.0),
                        child: ListTile(
                          title: Text(todo.task),
                          subtitle: Text(DateFormat("dd-MM-yyyy h:mm
a").format(todo.createdOn.toDate())),
                          trailing: Checkbox(
                            value: todo.isDone,
                            onChanged: (value) {
                              Todos updatedTodo = todo.copyWith(
                                isDone: value!,
                                updatedOn: Timestamp.now(),
                              );
                              _databaseService.updateTodo(id, updatedTodo);
                            },
                          ),
                          onLongPress: () => _databaseService.deleteTodo(id)
                        ),
                      );
                    },
                  );
                }
              ),
            ),
          ],
        ),
      ),
    );
  }
}
```

# Mobile Application Development Lab

# CSL 341

# Lab Journal 11

## Teacher: Muhammad Athesham Noor



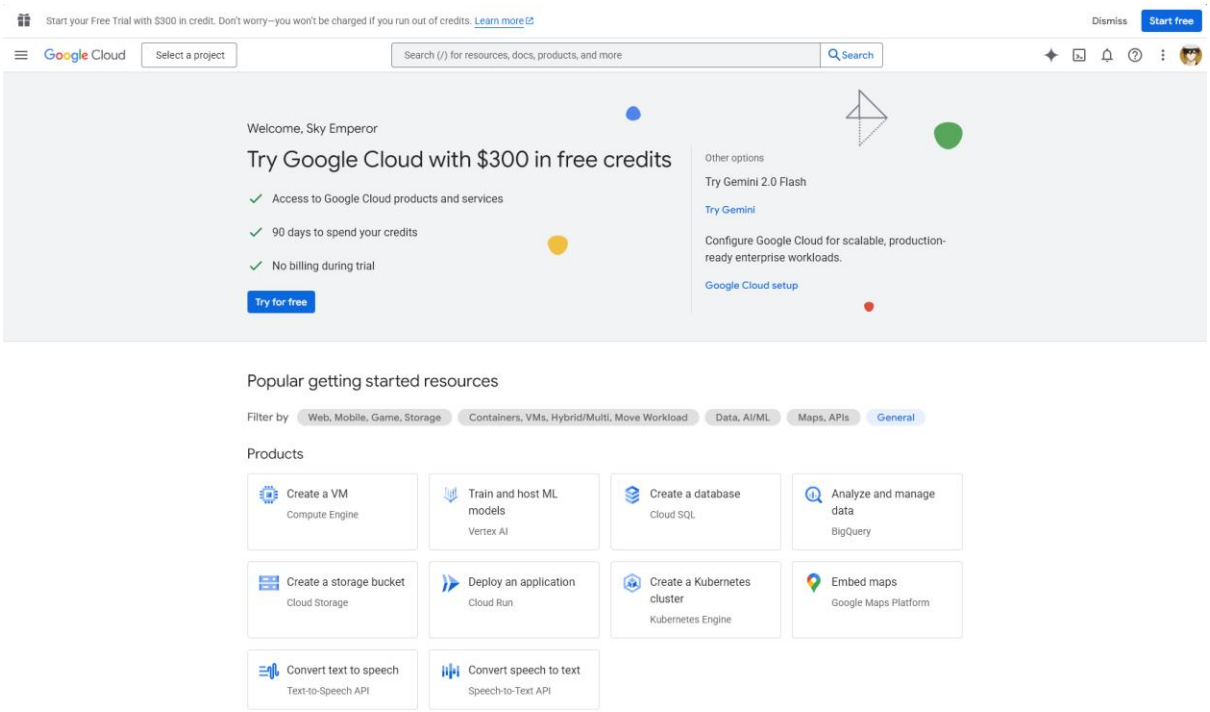

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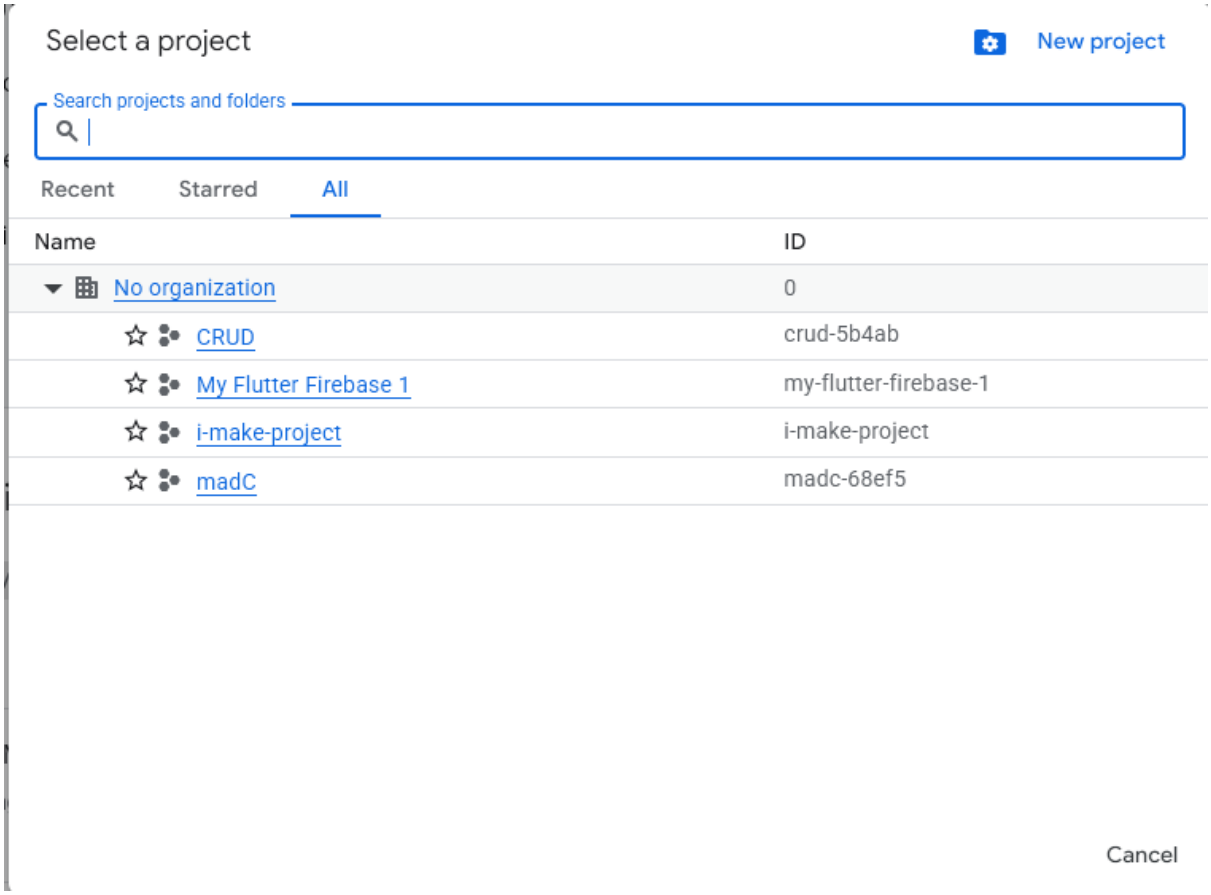
**Question:**

Integrate Google Maps API in flutter and place markers on the maps. Placing markers within 5km of each other must trigger a message that informs about the proximity. Place a button that submits the current location information to Firebase Firestore.

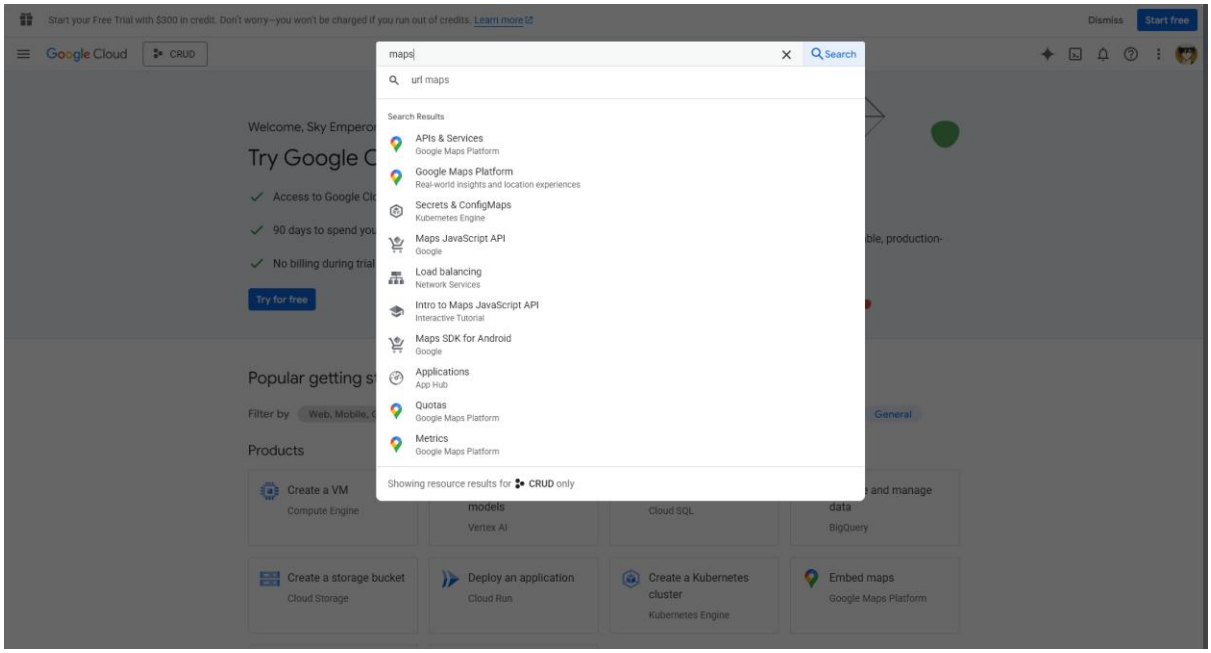
- Go to console.cloud.google.com



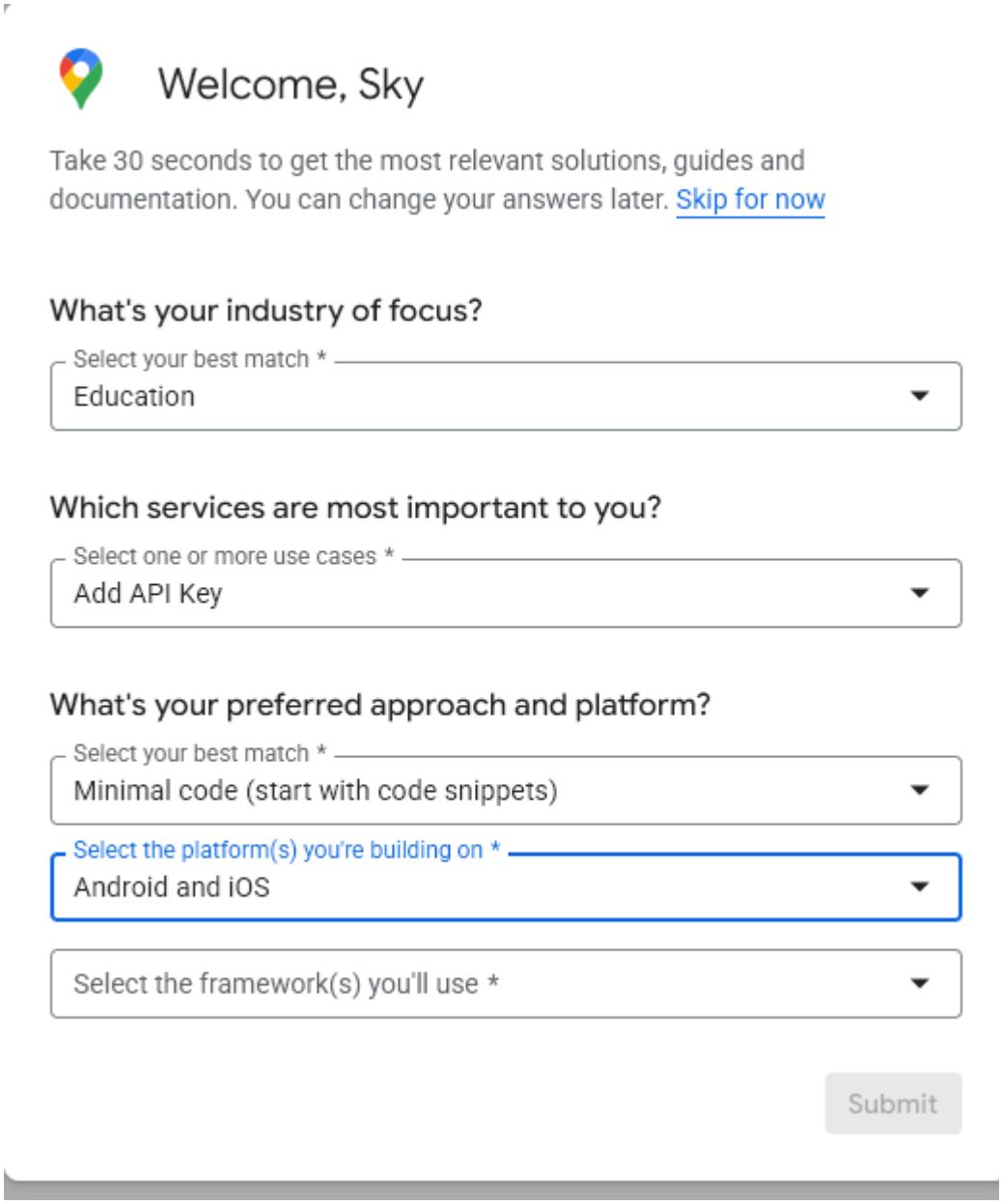

- Create a project or select an existing one.

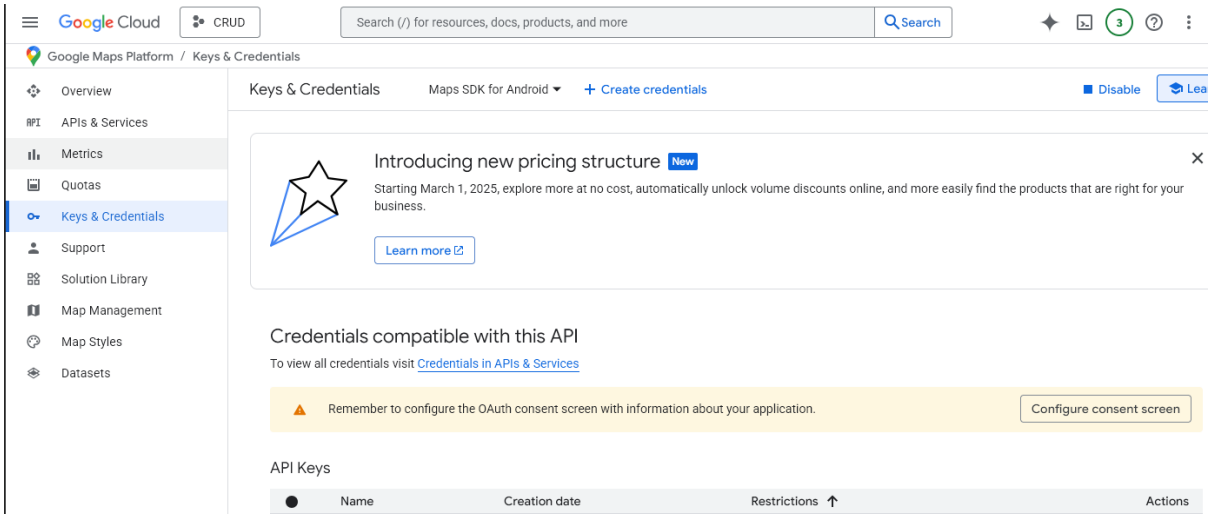



- Search for maps in the search bar after selecting the project



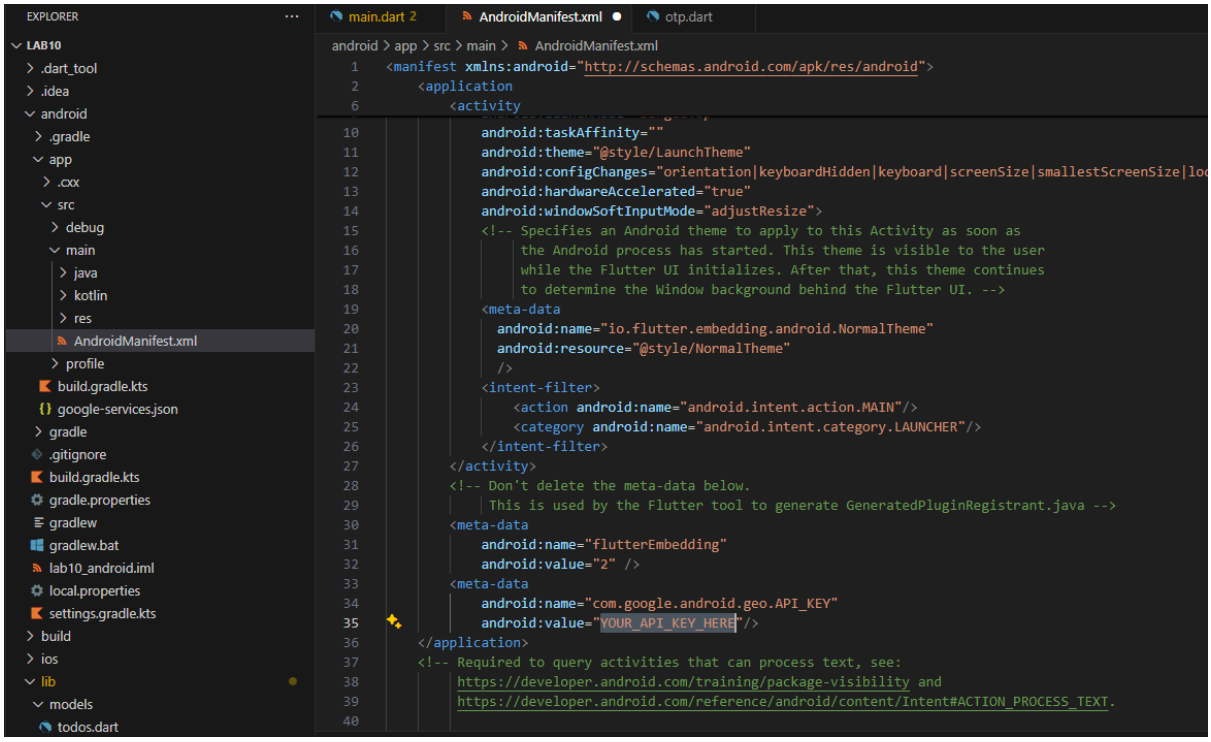

- Enter your payment information




- 
- Go to keys and credentials in the google maps api area and create credentials to generate your key
- 

- Go to android/app/src/main/AndroidManifest.xml and paste while putting your api key

```
<meta-data
            android:name="com.google.android.geo.API_KEY"
            android:value="YOUR_API_KEY_HERE"/>
```




- Then paste this

```
    <uses-permission android:name="android.permission.INTERNET"/>
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>
```

```
        In particular, this is used by the Flutter engine in io.flutter.plugin.text.Pr
    <queries>
        <intent>
            <action android:name="android.intent.action.PROCESS_TEXT"/>
            <data android:mimeType="text/plain"/>
        </intent>
    </queries>
    <uses-permission android:name="android.permission.INTERNET"/>
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>
</manifest>
```

- Run this command flutter pub add google_maps_flutter

Code:

```
import 'package:flutter/material.dart';
import 'package:google_maps_flutter/google_maps_flutter.dart';
import 'package:cloud_firestore/cloud_firestore.dart';
import 'package:firebase_core/firebase_core.dart';
import 'dart:math';
```

```dart
void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp();
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(home: MapSample());
  }
}

class MapSample extends StatefulWidget {
  @override
  State<MapSample> createState() => MapSampleState();
}

class MapSampleState extends State<MapSample> {
  GoogleMapController? mapController;
  late final CollectionReference location;
  Marker? _origin;
  Marker? _destination;
  late final BitmapDescriptor _originIcon;
  late final BitmapDescriptor _destinationIcon;

  @override
  void initState() {
    super.initState();
    location = FirebaseFirestore.instance.collection("location");
    aFunction();
  }

  void aFunction() async {
    _originIcon = await BitmapDescriptor.asset(
      ImageConfiguration(size: Size(50, 50)),
      "assets/desserts.png",
    );
    _destinationIcon = await BitmapDescriptor.asset(
      ImageConfiguration(size: Size(50, 50)),
      "assets/fastfood.png",
    );
  }

  void _submitLocation() async {
    if (mapController != null) {
      LatLng latLng = await mapController!.getLatLng(
```

```
        ScreenCoordinate(x: 200, y: 400),
      );
      await location.add({
        "latitude": latLng.latitude,
        "longitude": latLng.longitude,
      });
    } else {
      print("MapController is not initialized yet.");
    }
  }

  double _calculateDistance(
    double lat1,
    double lon1,
    double lat2,
    double lon2,
  ) {
    const p = 0.017453292519943295; // Pi/180
    final a =
        0.5 -
        (cos((lat2 - lat1) * p) / 2) +
        cos(lat1 * p) * cos(lat2 * p) * (1 - cos((lon2 - lon1) * p)) / 2;

    return 12742 * asin(sqrt(a)); // Distance in km
  }

  void dispose() {
    mapController?.dispose();
    super.dispose();
  }

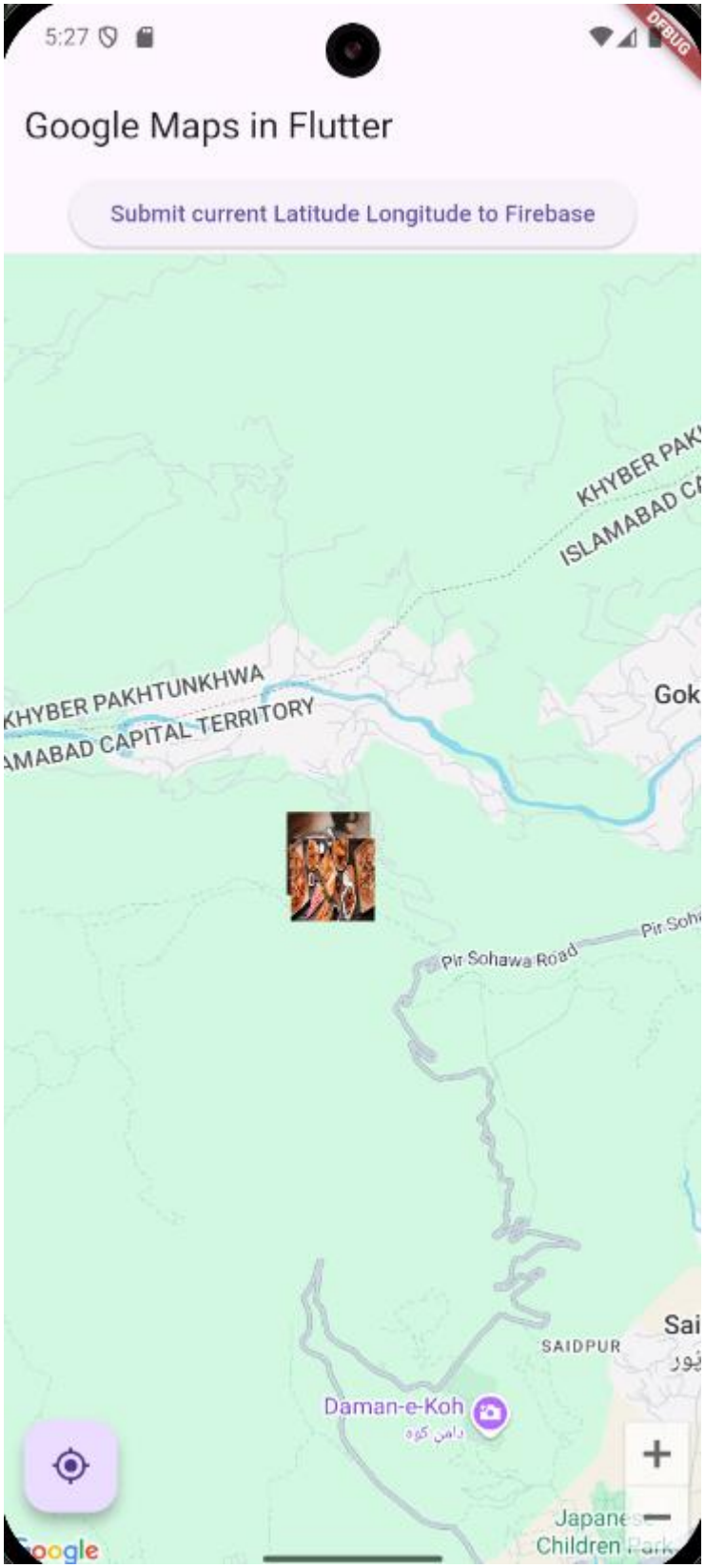
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: Text('Google Maps in Flutter')),
      floatingActionButtonLocation: FloatingActionButtonLocation.startFloat,
      floatingActionButton: FloatingActionButton(
        onPressed: () {
          mapController?.animateCamera(
            CameraUpdate.newCameraPosition(
              CameraPosition(target: LatLng(33.6844, 73.0479), zoom: 11.0),
            ),
          );
        },
        child: const Icon(Icons.my_location),
      ),
      body: Column(
        children: [
```

```
          ElevatedButton(
            onPressed: _submitLocation,
            child: const Text("Submit current Latitude Longitude to
Firebase"),
          ),
          Expanded(
            child: GoogleMap(
              zoomControlsEnabled: true,
              markers: {
                if (_origin != null) _origin!,
                if (_destination != null) _destination!,
              },
              onLongPress: (LatLng location) {
                if (_origin == null) {
                  setState(() {
                    _origin = Marker(
                      markerId: const MarkerId("origin"),
                      position: location,
                      infoWindow: const InfoWindow(title: "Origin"),
                      draggable: true,
                      icon: _originIcon,
                    );
                  });
                } else if (_destination == null) {
                  setState(() {
                    _destination = Marker(
                      markerId: const MarkerId("destination"),
                      position: location,
                      infoWindow: const InfoWindow(title: "Destination"),
                      draggable: true,
                      icon: _destinationIcon,
                    );

                    double distance = _calculateDistance(
                    _origin!.position.latitude,
                    _origin!.position.longitude,
                    _destination!.position.latitude,
                    _destination!.position.longitude,
                  );

                  if (distance <= 5) {
                    // 5 km radius
                    print("Origin and Destination are within 5 km!");
                  } else {
                    print("They are more than 5 km apart.");
                  }
                  });
```

```
                } else {
                  setState(() {
                    _origin = null;
                    _destination = null;
                  });
                }
              },
              onMapCreated: (GoogleMapController controller) {
                setState(() {
                  mapController = controller;
                });
              },
              initialCameraPosition: CameraPosition(
                target: LatLng(33.6844, 73.0479),
                zoom: 11.0,
              ),
            ),
          ),
        ],
      ),
    );
  }
}
```


5:27
DEBUG
Google Maps in Flutter
Submit current Latitude Longitude to Firebase
KHYBER PAKHTUNKHWA
AMABAD CAPITAL TERRITORY
Pir Sohawa Road
SAIDPUR
Daman-e-Koh
Google

# Mobile Application Development Lab

# CSL 341

# Lab Journal 12

## Teacher: Muhammad Athesham Noor




Department of Computer Science

BAHRIA UNIVERSITY, ISLAMABAD

**Question:**

Make a flutter app that authenticates a user using firebase phone number login method.

Main.dart:

```dart
import 'package:firebase_auth/firebase_auth.dart';

import 'package:firebase_core/firebase_core.dart';
import 'package:flutter/material.dart';
import 'home_screen.dart';
import 'phone_auth_screen.dart';
import 'package:provider/provider.dart';
import 'auth_service.dart';

void main() async {
  WidgetsFlutterBinding.ensureInitialized();
  await Firebase.initializeApp();
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MultiProvider(
      providers: [
        Provider<AuthService>(create: (_) => AuthService()),
      ],
      child: MaterialApp(
        title: 'Phone Auth Demo',
        theme: ThemeData(primarySwatch: Colors.blue),
        home: AuthWrapper(),
      ),
    );
  }
}

class AuthWrapper extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    final authService = Provider.of<AuthService>(context);
    return StreamBuilder<User?>(
      stream: authService.authStateChanges,
      builder: (context, snapshot) {
        if (snapshot.connectionState == ConnectionState.active) {
          final user = snapshot.data;
          return user == null ? PhoneAuthScreen() : HomeScreen();
        }
```

```
        return Scaffold(body: Center(child: CircularProgressIndicator()));
      },
    );
  }
}
```

Home_screen.dart:

```
import 'package:firebase_auth/firebase_auth.dart';
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
import 'auth_service.dart';

class HomeScreen extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    final authService = Provider.of<AuthService>(context);
    final userStream = authService.authStateChanges;

    return Scaffold(
      appBar: AppBar(
        title: Text('Home'),
        actions: [
          IconButton(
            icon: Icon(Icons.logout),
            onPressed: () async {
              await authService.signOut();
            },
          ),
        ],
      ),
      body: Center(
        child: StreamBuilder<User?>(
          stream: userStream,
          builder: (context, snapshot) {
            if (snapshot.hasData) {
              final user = snapshot.data;
              return Column(
                mainAxisAlignment: MainAxisAlignment.center,
                children: [
                  Text('Logged in as:'),
                  Text(user?.phoneNumber ?? 'No phone number'),
                ],
              );
            }
            return CircularProgressIndicator();
          },
```

```
        ),
      ),
    );
  }
}
```

Phone_auth_screen.dart:

```dart
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
import 'auth_service.dart';

class PhoneAuthScreen extends StatefulWidget {
  @override
  _PhoneAuthScreenState createState() => _PhoneAuthScreenState();
}

class _PhoneAuthScreenState extends State<PhoneAuthScreen> {
  final TextEditingController _phoneController = TextEditingController();
  final TextEditingController _otpController = TextEditingController();

  String? _verificationId;
  bool _isCodeSent = false;
  bool _isLoading = false;

  @override
  Widget build(BuildContext context) {
    final authService = Provider.of<AuthService>(context);

    return Scaffold(
      appBar: AppBar(title: Text('Phone Auth')),
      body: Padding(
        padding: const EdgeInsets.all(20.0),
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            if (!_isCodeSent) ...[
              TextField(
                controller: _phoneController,
                decoration: InputDecoration(
                  labelText: 'Phone Number',
                  hintText: '+1234567890',
                  border: OutlineInputBorder(),
                ),
                keyboardType: TextInputType.phone,
              ),
              SizedBox(height: 20),
```

```
            _isLoading
                ? CircularProgressIndicator()
                : ElevatedButton(
                    onPressed: () async {
                      setState(() => _isLoading = true);
                      await authService.verifyPhoneNumber(
                        _phoneController.text,
                        onCodeSent: (verificationId) {
                          setState(() {
                            _verificationId = verificationId;
                            _isCodeSent = true;
                            _isLoading = false;
                          });
                        },
                        onVerificationFailed: (error) {
                          setState(() => _isLoading = false);
                          ScaffoldMessenger.of(context).showSnackBar(
                            SnackBar(content: Text(error.message ??
'Verification failed')),
                          );
                        },
                        onVerificationCompleted: (credential) async {
                          await authService.signInWithPhoneNumber(
                            credential.verificationId ?? '',
                            credential.smsCode ?? '',
                          );
                        },
                      );
                    },
                    child: Text('Send OTP'),
                  ),
          ] else ...[
            Text('Enter OTP sent to ${_phoneController.text}'),
            SizedBox(height: 20),
            TextField(
              controller: _otpController,
              decoration: InputDecoration(
                labelText: 'OTP',
                border: OutlineInputBorder(),
              ),
              keyboardType: TextInputType.number,
            ),
            SizedBox(height: 20),
            _isLoading
                ? CircularProgressIndicator()
                : ElevatedButton(
                    onPressed: () async {
                      setState(() => _isLoading = true);
```

```
                        final user = await authService.signInWithPhoneNumber(
                          _verificationId!,
                          _otpController.text,
                        );
                        if (user == null) {
                          setState(() => _isLoading = false);
                          ScaffoldMessenger.of(context).showSnackBar(
                            SnackBar(content: Text('Invalid OTP')),
                          );
                        }
                      },
                      child: Text('Verify OTP'),
                    ),
              ],
            ],
          ),
        ),
      );
  }
}
```

Auth_service.dart:

```
import 'package:firebase_auth/firebase_auth.dart';


class AuthService {
  final FirebaseAuth _auth = FirebaseAuth.instance;

  Stream<User?> get authStateChanges => _auth.authStateChanges();

  Future<void> verifyPhoneNumber(
    String phoneNumber, {
    required Function(String) onCodeSent,
    required Function(FirebaseAuthException) onVerificationFailed,
    required Function(PhoneAuthCredential) onVerificationCompleted,
  }) async {
    await _auth.verifyPhoneNumber(
      phoneNumber: phoneNumber,
      verificationCompleted: onVerificationCompleted,
      verificationFailed: onVerificationFailed,
      codeSent: (verificationId, _) => onCodeSent(verificationId),
      codeAutoRetrievalTimeout: (verificationId) {},
      timeout: const Duration(seconds: 60),
    );
  }

  Future<User?> signInWithPhoneNumber(
      String verificationId, String smsCode) async {
```

```
    try {
      final credential = PhoneAuthProvider.credential(
        verificationId: verificationId,
        smsCode: smsCode,
      );
      final userCredential = await _auth.signInWithCredential(credential);
      return userCredential.user;
    } catch (e) {
      print("Error signing in: $e");
      return null;
    }
  }

  Future<void> signOut() async {
    await _auth.signOut();
  }
}
```


6:27
DEBUG
Phone Auth
Phone Number
Send OTP

# Mobile Application Development Lab

## CSL 341

## Lab Journal 13

## Teacher: Muhammad Athesham Noor




Department of Computer Science

BAHRIA UNIVERSITY, ISLAMABAD

**Question:**

Use FastAPI to perform CRUD operations.

User.dart:

```dart
class User {
  int? id;
  String name;
  String email;

  User({this.id, required this.name, required this.email});

  factory User.fromJson(Map<String, dynamic> json) => User(
    id: json['id'],
    name: json['name'],
    email: json['email'],
  );

  Map<String, dynamic> toJson() => {
    'name': name,
    'email': email,
  };
}
```

Main.dart:

```dart
import 'package:flutter/material.dart';
import 'package:api_project/user.dart';
import 'package:api_project/api_service.dart';

void main() {
    runApp(MyApp());
}

class MyApp extends StatelessWidget {
    final ApiService apiService = ApiService();

    @override
    Widget build(BuildContext context) {
        return MaterialApp(
            title: 'User List',
            home: UserListScreen(apiService: apiService),
        );
    }
}
```

```dart
class UserListScreen extends StatefulWidget {
    final ApiService apiService;

    UserListScreen({required this.apiService});

    @override
    State<UserListScreen> createState() => _UserListScreenState();
}

class _UserListScreenState extends State<UserListScreen> {
    late Future<List<User>> _usersFuture;

    @override
    void initState() {
        super.initState();
        _refreshUsers();
    }

    void _refreshUsers() {
        setState(() {
            _usersFuture = widget.apiService.getUsers();
        });
    }

    void _showUserDialog({User? user}) {
        final _nameController = TextEditingController(text: user?.name ?? '');
        final _emailController = TextEditingController(text: user?.email ??
'');

        showDialog(
            context: context,
            builder: (context) => AlertDialog(
                title: Text(user == null ? 'Add User' : 'Edit User'),
                content: Column(
                    mainAxisSize: MainAxisSize.min,
                    children: [
                        TextField(
                            controller: _nameController,
                            decoration: InputDecoration(labelText: 'Name'),
                        ),
                        TextField(
                            controller: _emailController,
                            decoration: InputDecoration(labelText: 'Email'),
                        ),
                    ],
                ),
                actions: [
                    TextButton(
```

```
                        onPressed: () => Navigator.pop(context),
                        child: Text('Cancel'),
                    ),
                    ElevatedButton(
                        onPressed: () async {
                            String name = _nameController.text.trim();
                            String email = _emailController.text.trim();
                            if (name.isEmpty || email.isEmpty) return;
                            if (user == null) {
                                int newId =
DateTime.now().millisecondsSinceEpoch;
                                await widget.apiService.createUser(
                                    User(id: newId, name: name, email: email),
                                );
                            } else {
                                user.name = name;
                                user.email = email;
                                await widget.apiService.updateUser(user.id!,
user);
                            }
                            Navigator.pop(context);
                            _refreshUsers();
                        },
                        child: Text(user == null ? 'Add' : 'Save'),
                    ),
                ],
            ),
        );
    }

    void _deleteUser(int id) async {
        await widget.apiService.deleteUser(id);
        _refreshUsers();
    }

    @override
    Widget build(BuildContext context) {
        return Scaffold(
            appBar: AppBar(
                title: Text('Users'),
            ),
            body: FutureBuilder<List<User>>(
                future: _usersFuture,
                builder: (context, snapshot) {
                    if (snapshot.connectionState == ConnectionState.waiting)
                        return Center(child: CircularProgressIndicator());
                    if (!snapshot.hasData || snapshot.data!.isEmpty)
                        return Center(child: Text('No users found.'));
```

```
                    final users = snapshot.data!;
                    return ListView.builder(
                        itemCount: users.length,
                        itemBuilder: (context, index) {
                            final user = users[index];
                            return ListTile(
                                title: Text(user.name),
                                subtitle: Text(user.email),
                                trailing: Row(
                                    mainAxisSize: MainAxisSize.min,
                                    children: [
                                        IconButton(
                                            icon: Icon(Icons.edit),
                                            onPressed: () =>
_showUserDialog(user: user),
                                        ),
                                        IconButton(
                                            icon: Icon(Icons.delete),
                                            onPressed: () =>
_deleteUser(user.id!),
                                        ),
                                    ],
                                ),
                            );
                        },
                    );
                },
            ),
            floatingActionButton: FloatingActionButton(
                onPressed: () => _showUserDialog(),
                child: Icon(Icons.add),
            ),
        );
    }
}
```

Api_service.dart:

```
import 'dart:convert';
import 'package:http/http.dart' as http;
import 'user.dart';

class ApiService {
  final String baseUrl = "http://10.0.2.2:8000";

  Future<List<User>> getUsers() async {
    final response = await http.get(Uri.parse("$baseUrl/users/"));
    if (response.statusCode == 200) {
```

```
      Iterable list = json.decode(response.body);
      return list.map((json) => User.fromJson(json)).toList();
    } else {
      throw Exception("Failed to load users");
    }
  }

  Future<User> createUser(User user) async {
    final response = await http.post(
      Uri.parse("$baseUrl/users/"),
      headers: {"Content-Type": "application/json"},
      body: json.encode(user.toJson()),
    );
    if (response.statusCode == 200 || response.statusCode == 201) {
      return User.fromJson(json.decode(response.body));
    } else {
      throw Exception("Failed to create user");
    }
  }

  Future<void> updateUser(int id, User user) async {
    await http.put(
      Uri.parse("$baseUrl/users/$id"),
      headers: {"Content-Type": "application/json"},
      body: json.encode(user.toJson()),
    );
  }

  Future<void> deleteUser(int id) async {
    await http.delete(Uri.parse("$baseUrl/users/$id"));
  }
}
```

Crud.py:

```
from sqlalchemy.orm import Session
from models import User
from pydantic import BaseModel

class UserCreate(BaseModel):
    name: str
    email: str

def get_users(db: Session):
    return db.query(User).all()

def create_user(db: Session, user: UserCreate):
```

```
    db_user = User(name=user.name, email=user.email)
    db.add(db_user)
    db.commit()
    db.refresh(db_user)
    return db_user

def get_user(db: Session, user_id: int):
    return db.query(User).filter(User.id == user_id).first()

def update_user(db: Session, user_id: int, user: UserCreate):
    db_user = db.query(User).filter(User.id == user_id).first()
    db_user.name = user.name
    db_user.email = user.email
    db.commit()
    return db_user

def delete_user(db: Session, user_id: int):
    db_user = db.query(User).filter(User.id == user_id).first()
    db.delete(db_user)
    db.commit()
```

database.py:

```
from sqlalchemy import create_engine
from sqlalchemy.orm import sessionmaker, declarative_base

SQLALCHEMY_DATABASE_URL = "mysql+pymysql://root:@localhost:3306/students"

engine = create_engine(SQLALCHEMY_DATABASE_URL)
SessionLocal = sessionmaker(autocommit=False, autoflush=False, bind=engine)
Base = declarative_base()
```

Main.py:

```
from fastapi import FastAPI, Depends
from sqlalchemy.orm import Session
from database import Base, engine, SessionLocal
import crud, models
from pydantic import BaseModel
import joblib
import numpy as np
import os
#print(os.getcwd())


app = FastAPI()
```

```python
Base.metadata.create_all(bind=engine)

def get_db():
    db = SessionLocal()
    try:
        yield db
    finally:
        db.close()

class UserCreate(BaseModel):
    name: str
    email: str

@app.post("/users/")
def create_user(user: UserCreate, db: Session = Depends(get_db)):
    return crud.create_user(db, user)

@app.get("/users/")
def read_users(db: Session = Depends(get_db)):
    return crud.get_users(db)

@app.get("/users/{user_id}")
def read_user(user_id: int, db: Session = Depends(get_db)):
    return crud.get_user(db, user_id)

@app.put("/users/{user_id}")
def update_user(user_id: int, user: UserCreate, db: Session =
Depends(get_db)):
    return crud.update_user(db, user_id, user)

@app.delete("/users/{user_id}")
def delete_user(user_id: int, db: Session = Depends(get_db)):
    return crud.delete_user(db, user_id)

# Load model
model = joblib.load("linear_model.pkl")

# Define input schema
class PredictRequest(BaseModel):
    x: float

@app.post("/predict/")
def predict(data: PredictRequest):
    prediction = model.predict(np.array([[data.x]]))
    return {"input": data.x, "prediction": prediction[0]}
```

models.py:

```python
from sqlalchemy import Column, Integer, String
from database import Base

class User(Base):
    __tablename__ = "users"
    id = Column(Integer, primary_key=True, index=True)
    name = Column(String(50))
    email = Column(String(50), unique=True)
```

train.py:

```python
import numpy as np
import pandas as pd
from sklearn.linear_model import LinearRegression
import joblib

# Sample dummy data
data = {
    'x': np.arange(1, 11),
    'y': [3, 4, 2, 5, 6, 7, 8, 9, 10, 12]
}
df = pd.DataFrame(data)

# Train model
X = df[['x']]
y = df['y']
model = LinearRegression()
model.fit(X, y)

# Save model to file
joblib.dump(model, 'ahtesham.pkl')
print("Model saved as linear_model.pkl")
```